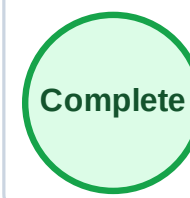


BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing

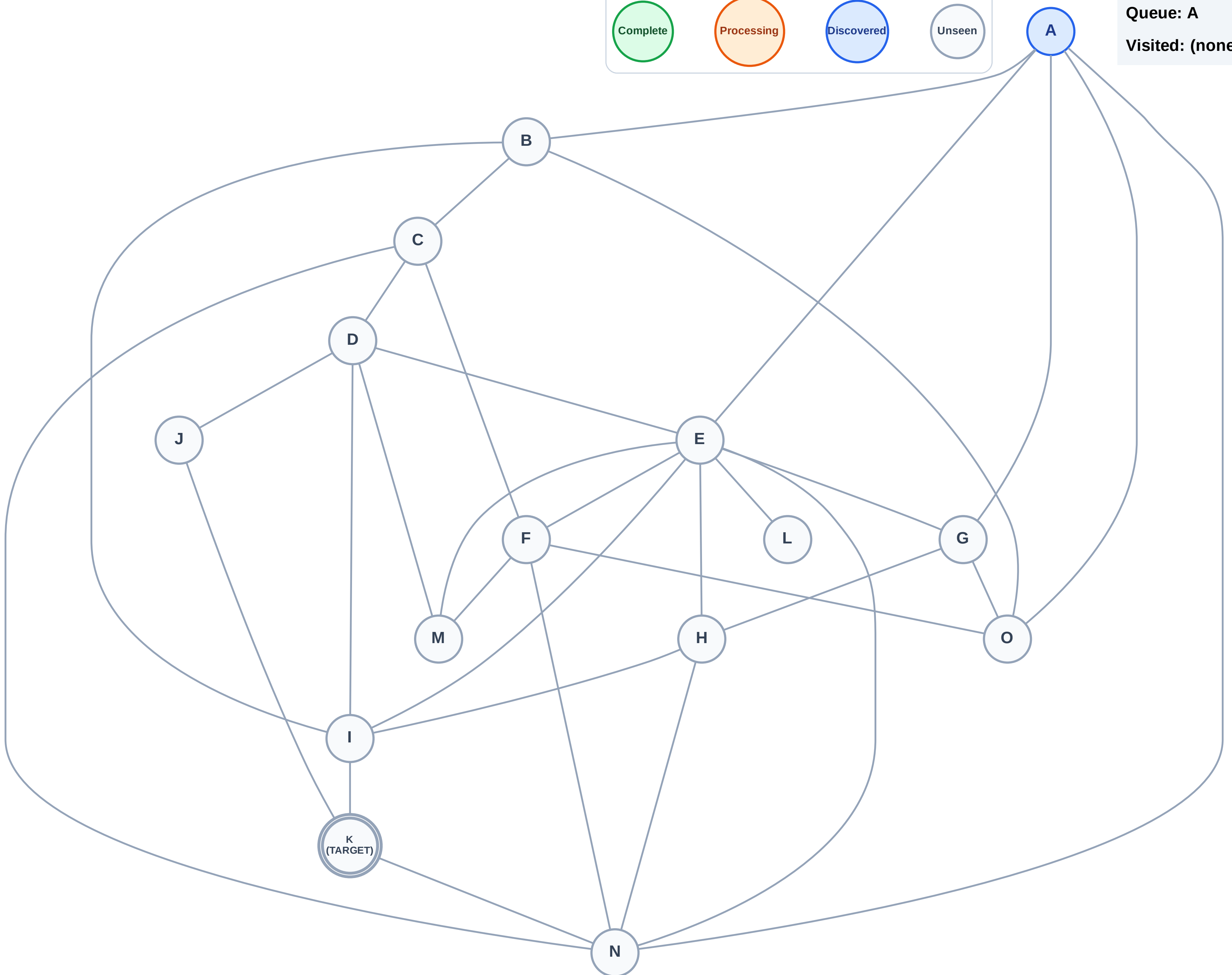


Discovered



Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

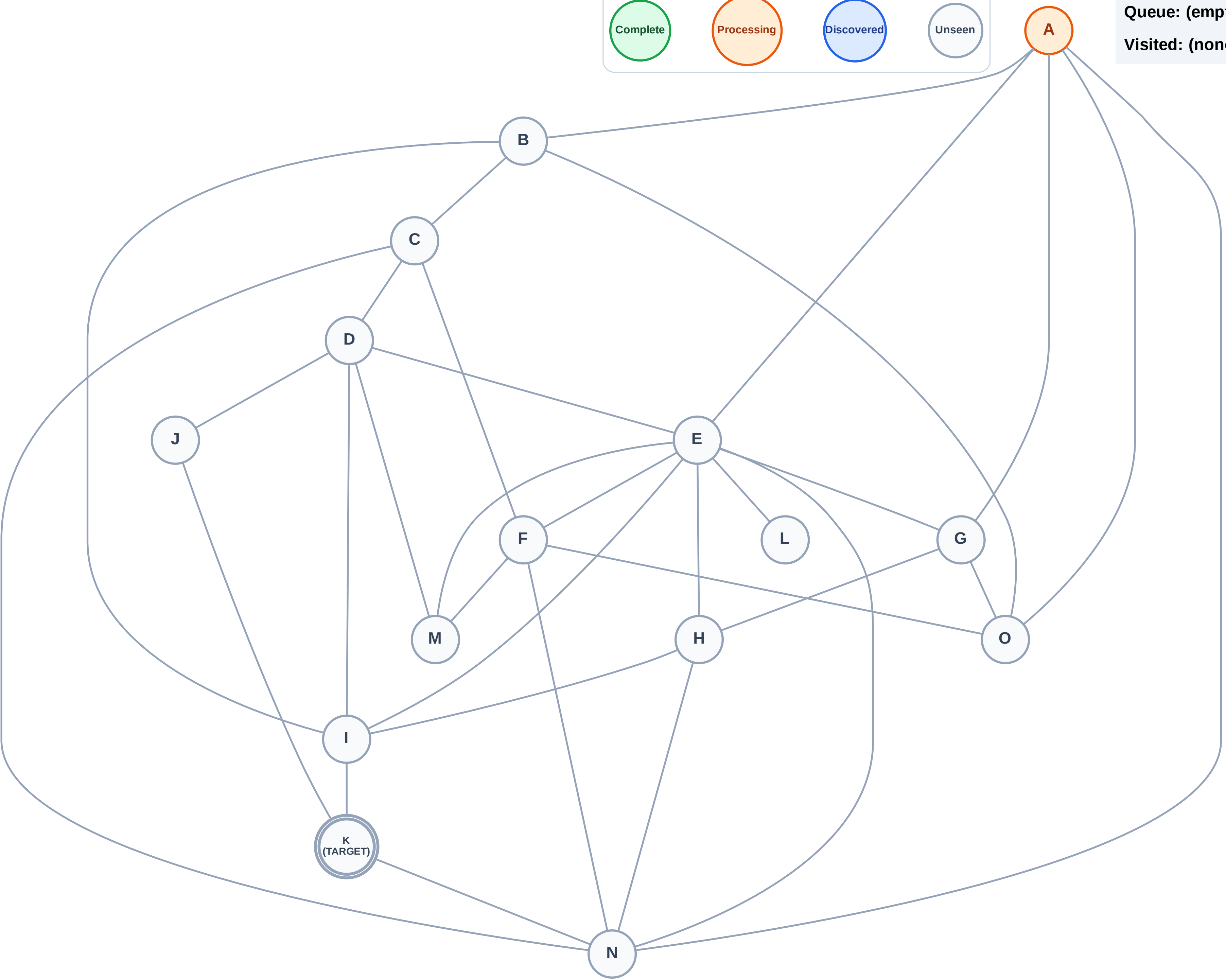
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

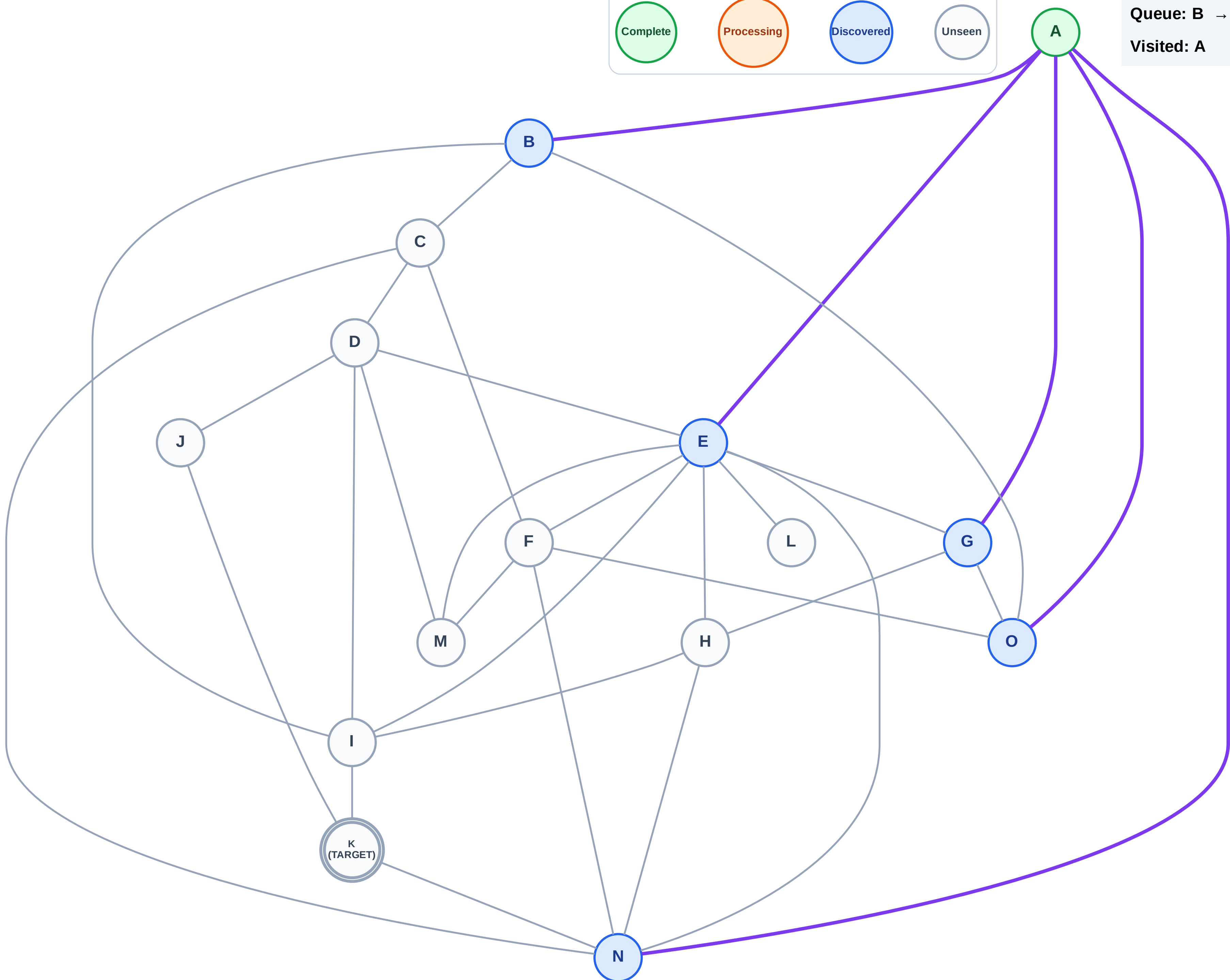
Step 3: Discover B, E, G, N, O from A

Legend



Queue: B → E → G → N → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

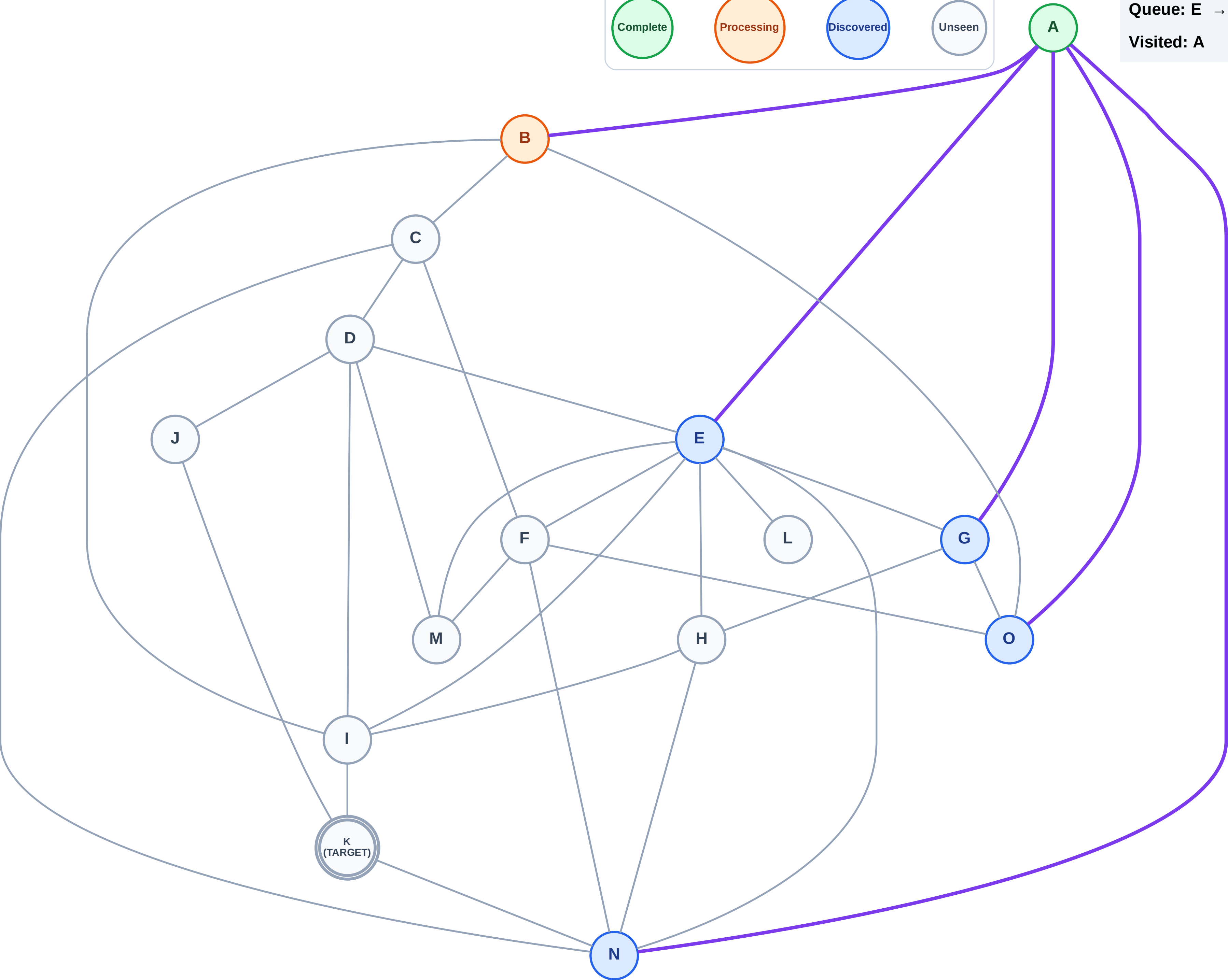
Processing

Discovered

Unseen

Queue: E → G → N → O

Visited: A



BFS Traversal

Step 5: Discover C, I from B

Legend



Complete



Processing



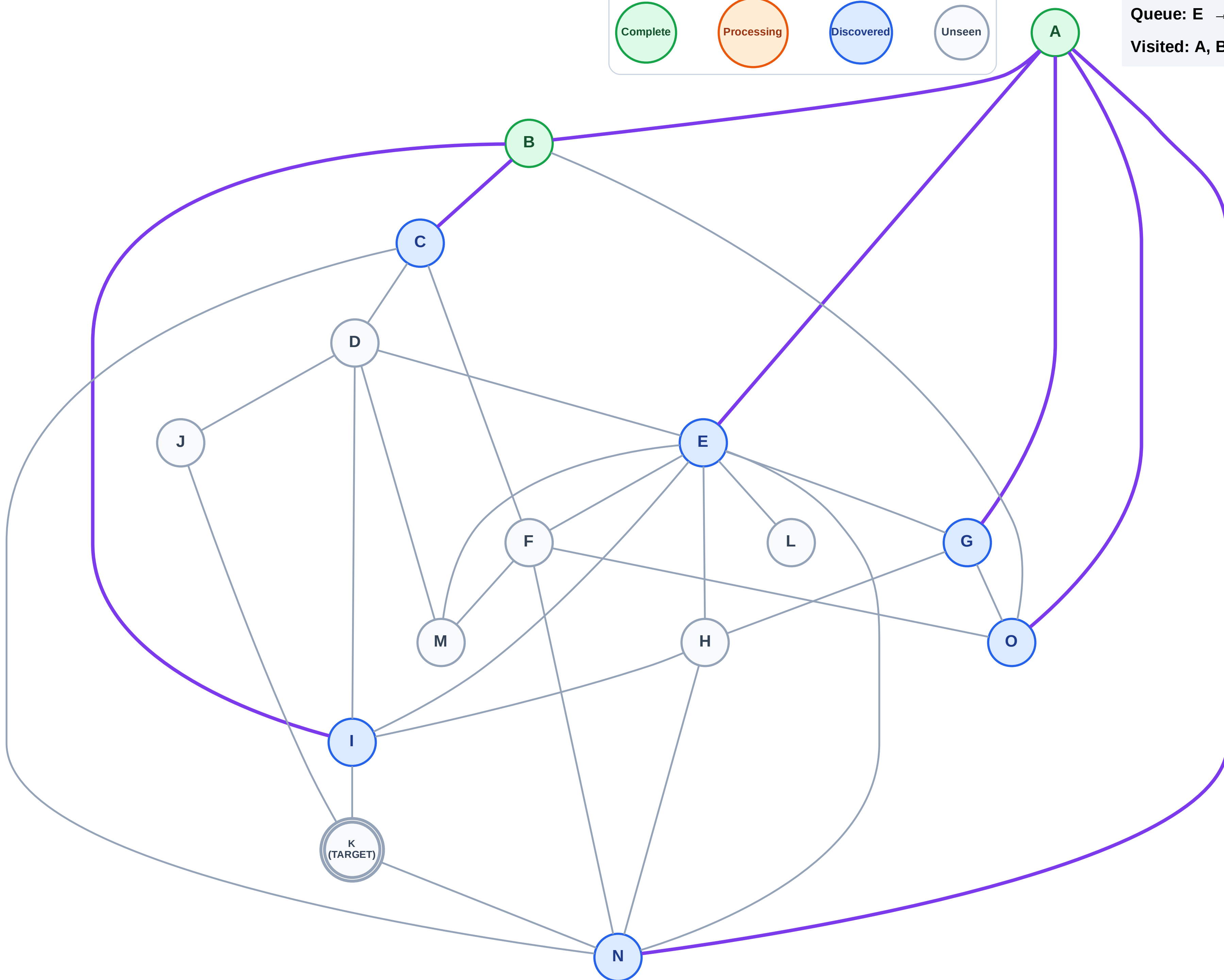
Discovered



Unseen

Queue: E → G → N → O → C → I

Visited: A, B



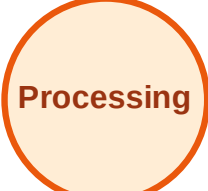
BFS Traversal

Step 6: Process node E

Legend



Complete



Processing



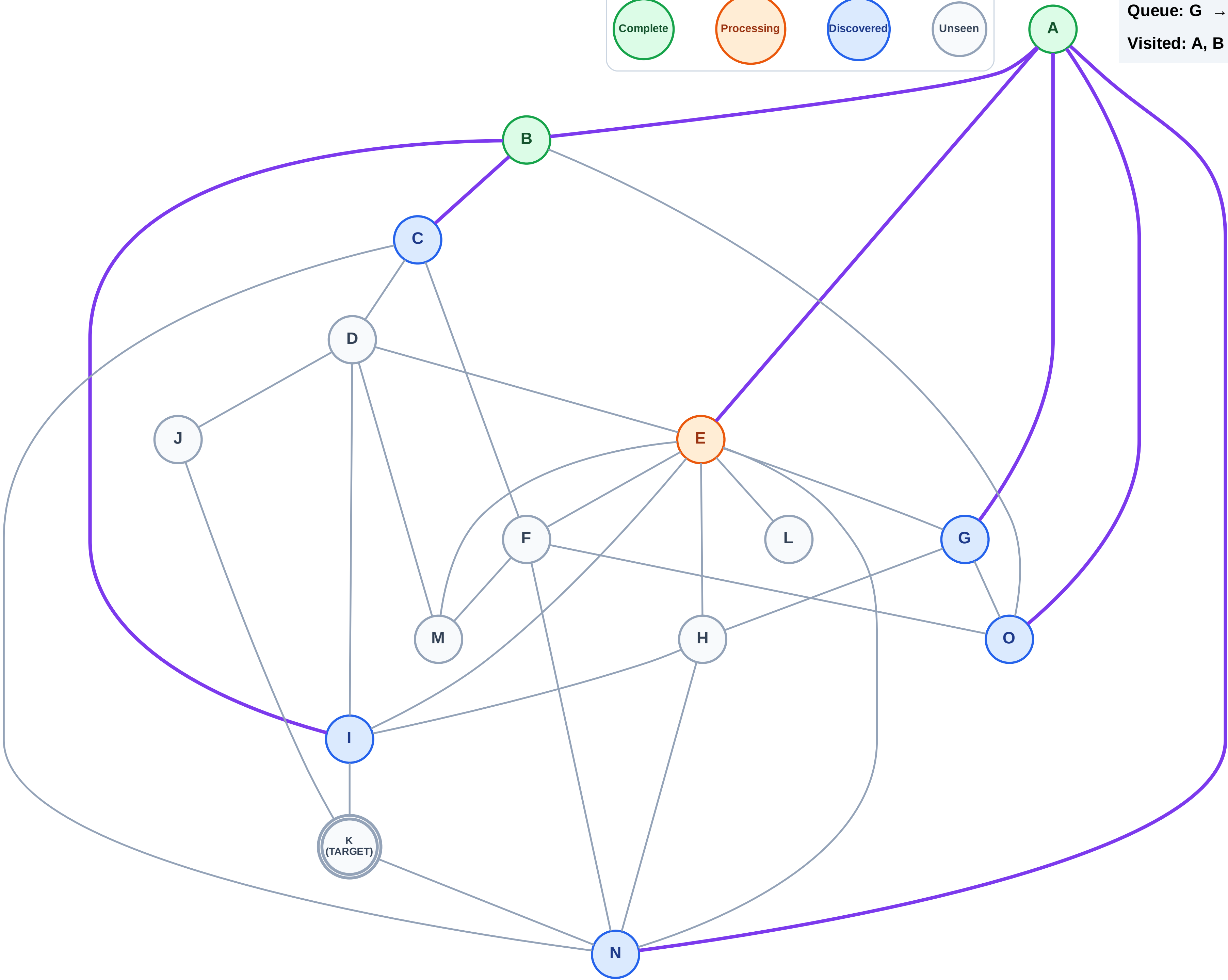
Discovered



Unseen

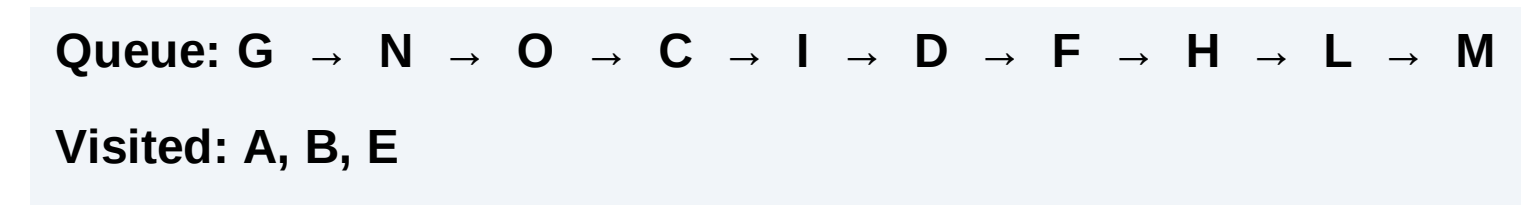
Queue: G → N → O → C → I

Visited: A, B



Step 7: Discover D, F, H, L, M from E

Step 7: Discover D, F, H, L, M from E



BFS Traversal
Step 8: Process node G

Legend

Complete

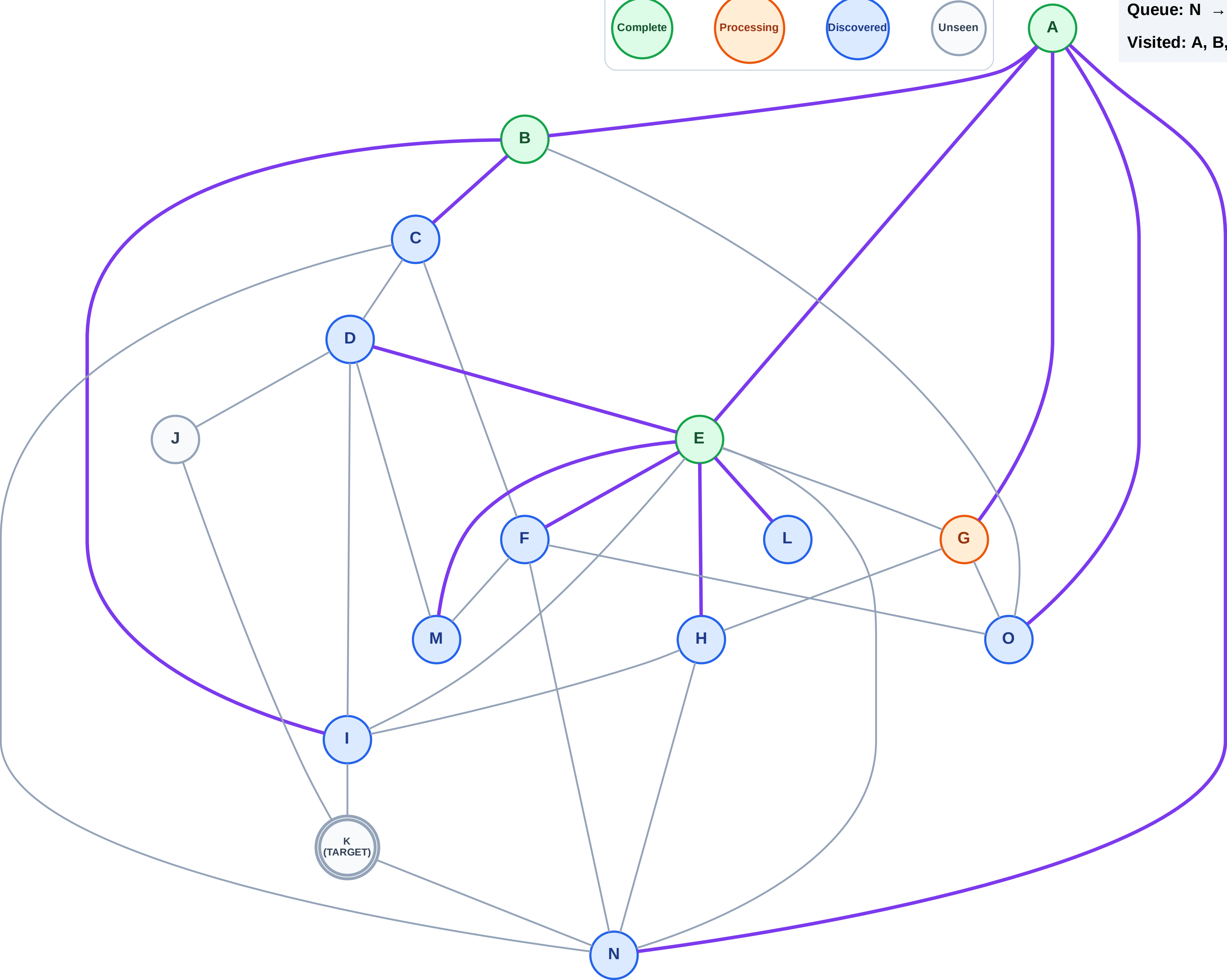
Processing

Discovered

Unseen

Queue: N → O → C → I → D → F → H → L → M

Visited: A, B, E



BFS Traversal

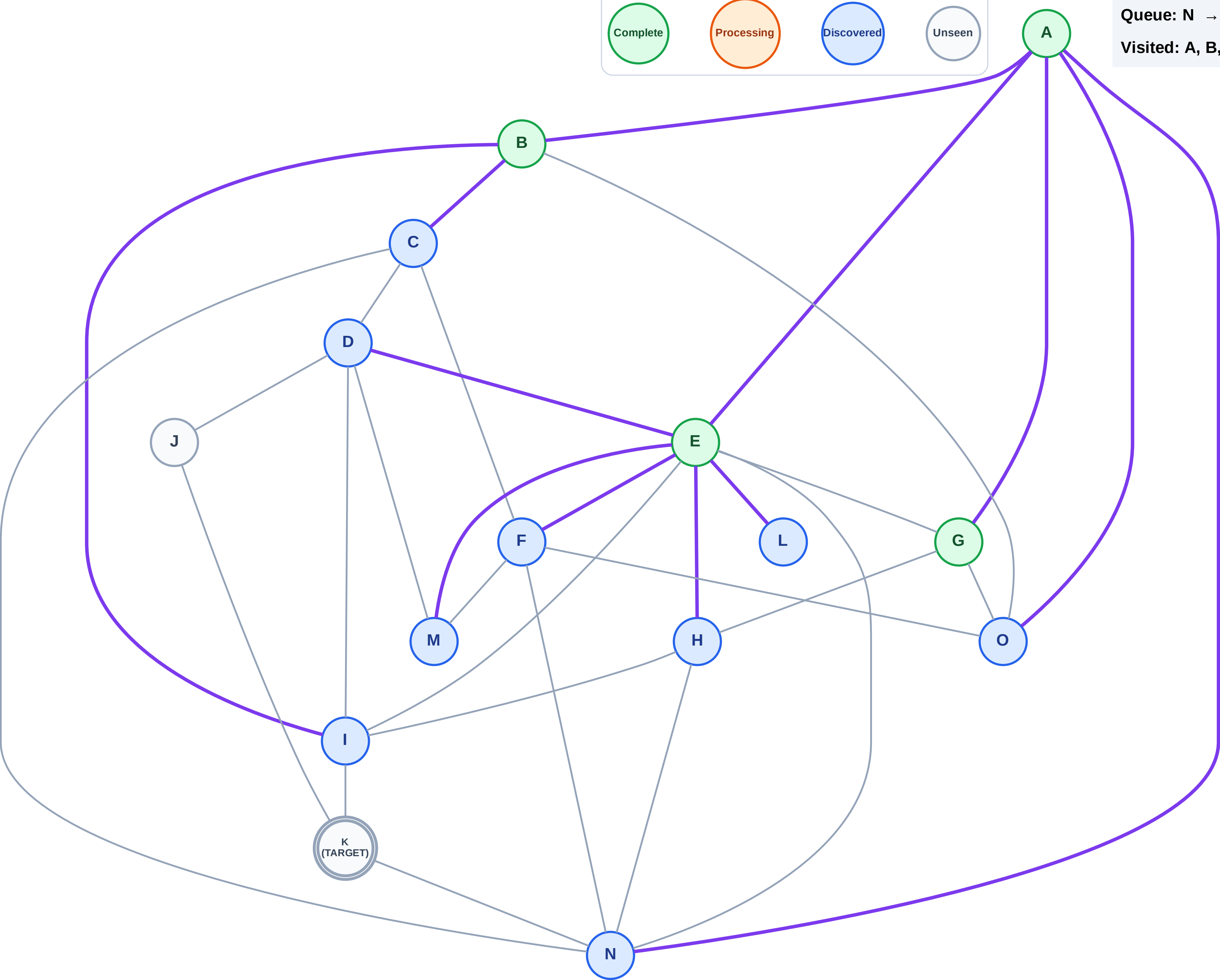
Step 9: No new neighbors from G

Legend



Queue: N → O → C → I → D → F → H → L → M

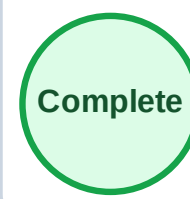
Visited: A, B, E, G



BFS Traversal

Step 10: Process node N

Legend



Complete



Processing

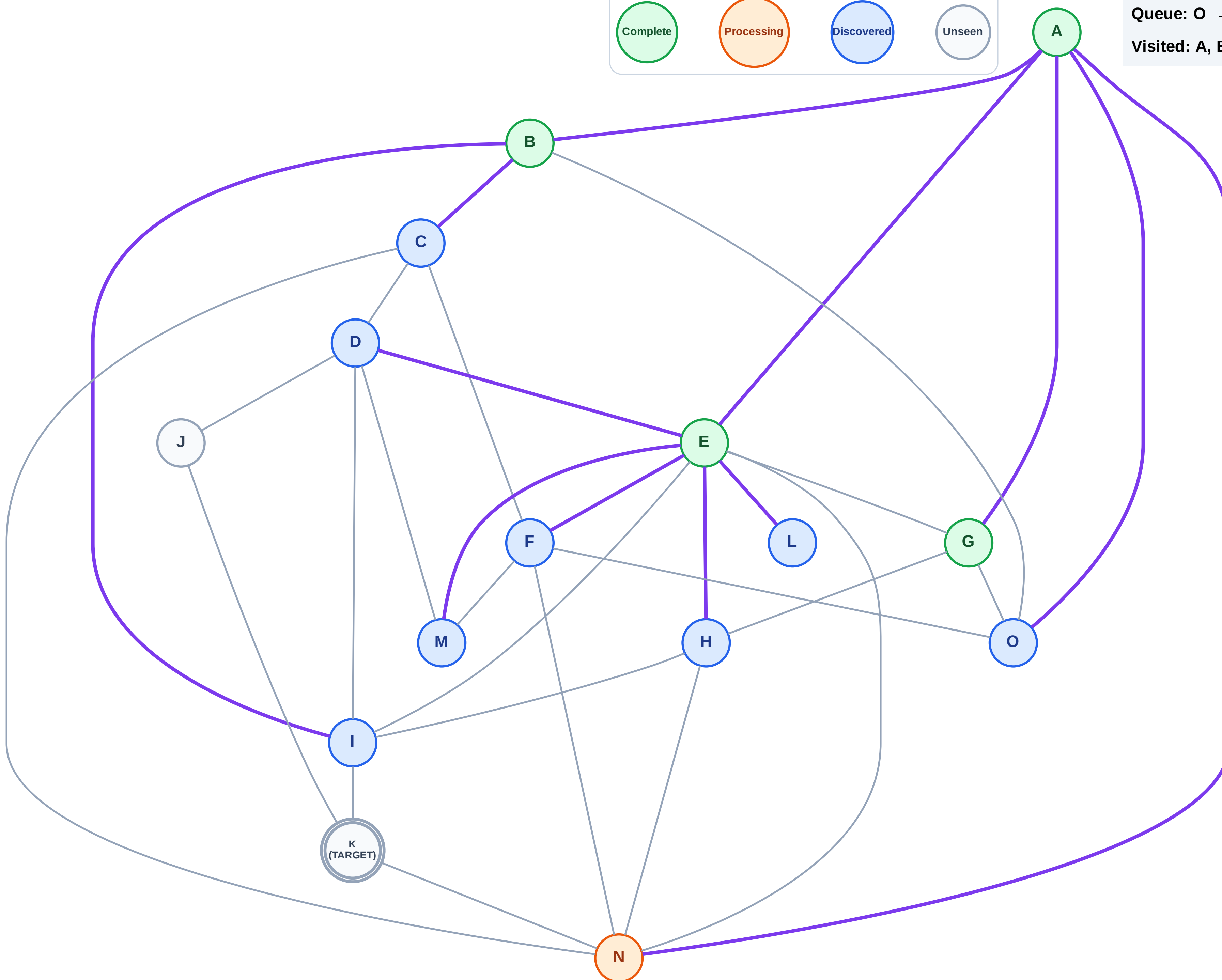


Discovered



Unseen

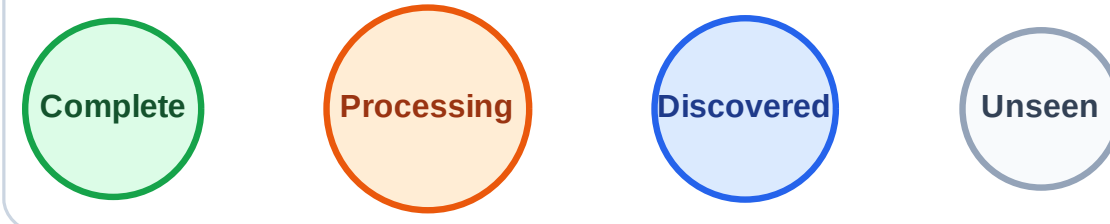
Queue: O → C → I → D → F → H → L → M
Visited: A, B, E, G



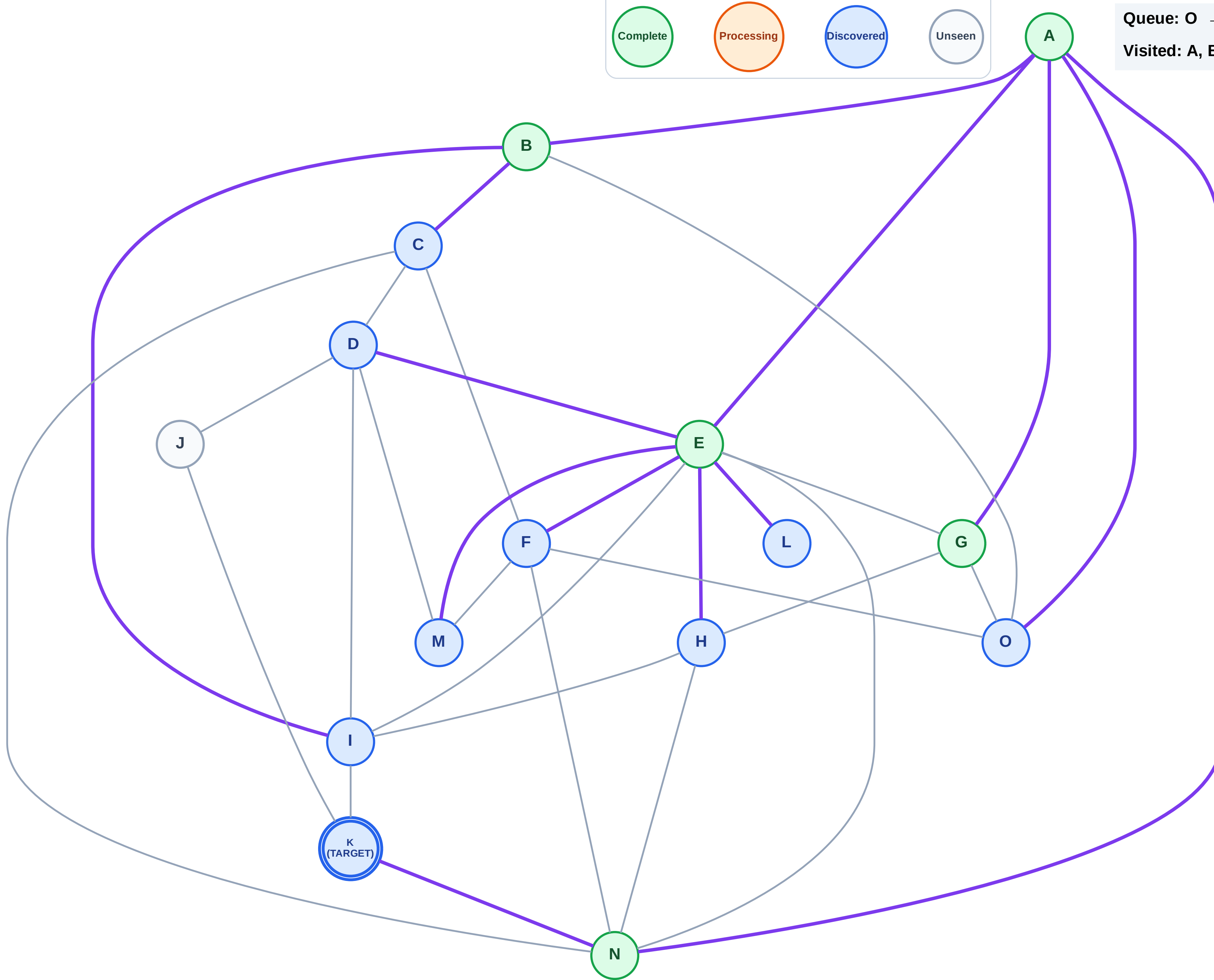
BFS Traversal

Step 11: Discover K from N

Legend



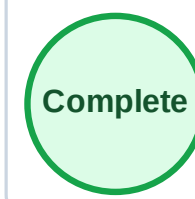
Queue: O → C → I → D → F → H → L → M → K
Visited: A, B, E, G, N



BFS Traversal

Step 12: Process node O

Legend



Complete



Processing



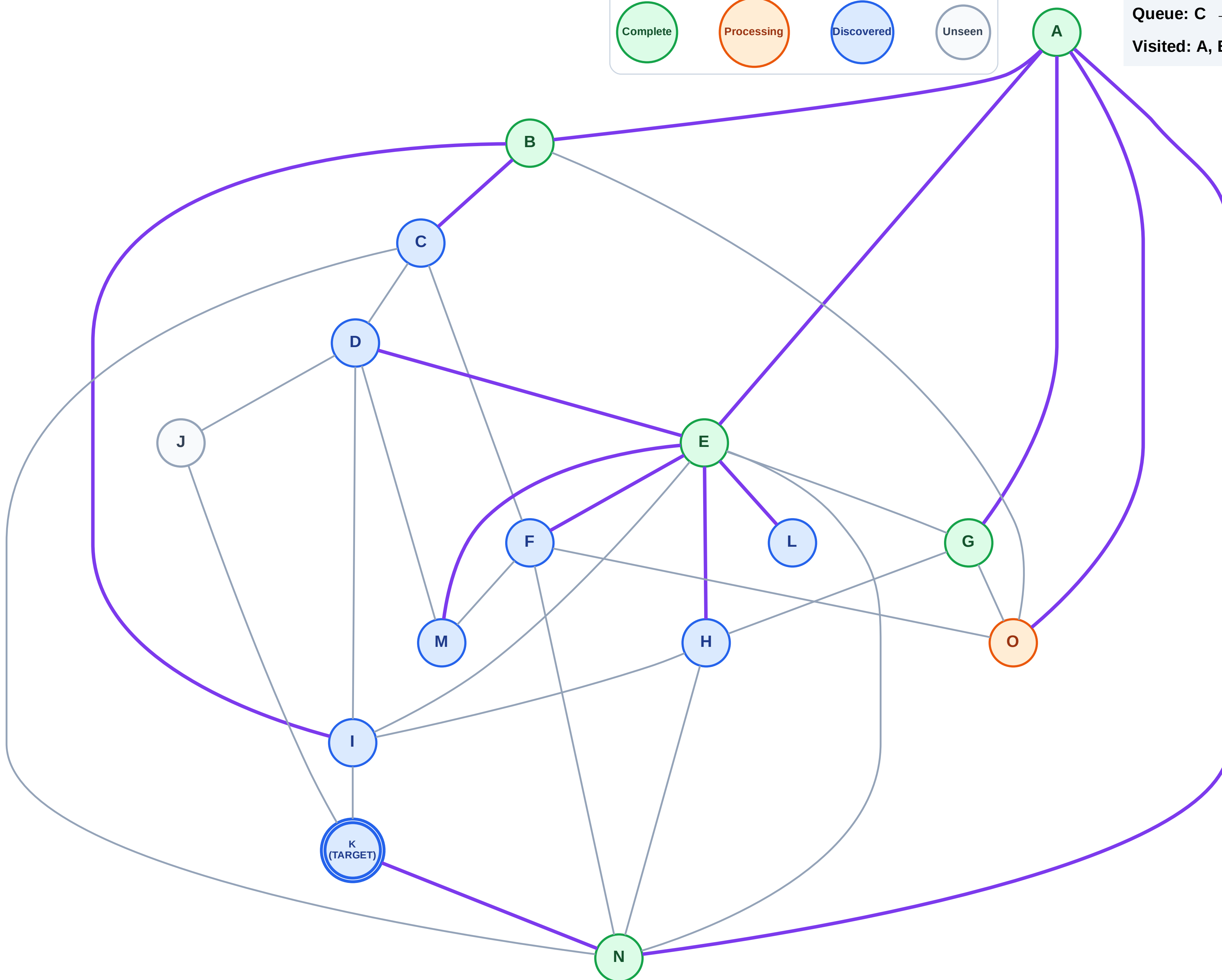
Discovered



Unseen

Queue: C → I → D → F → H → L → M → K

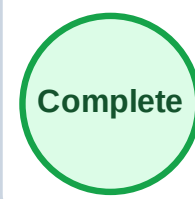
Visited: A, B, E, G, N



BFS Traversal

Step 13: No new neighbors from O

Legend



Complete



Processing

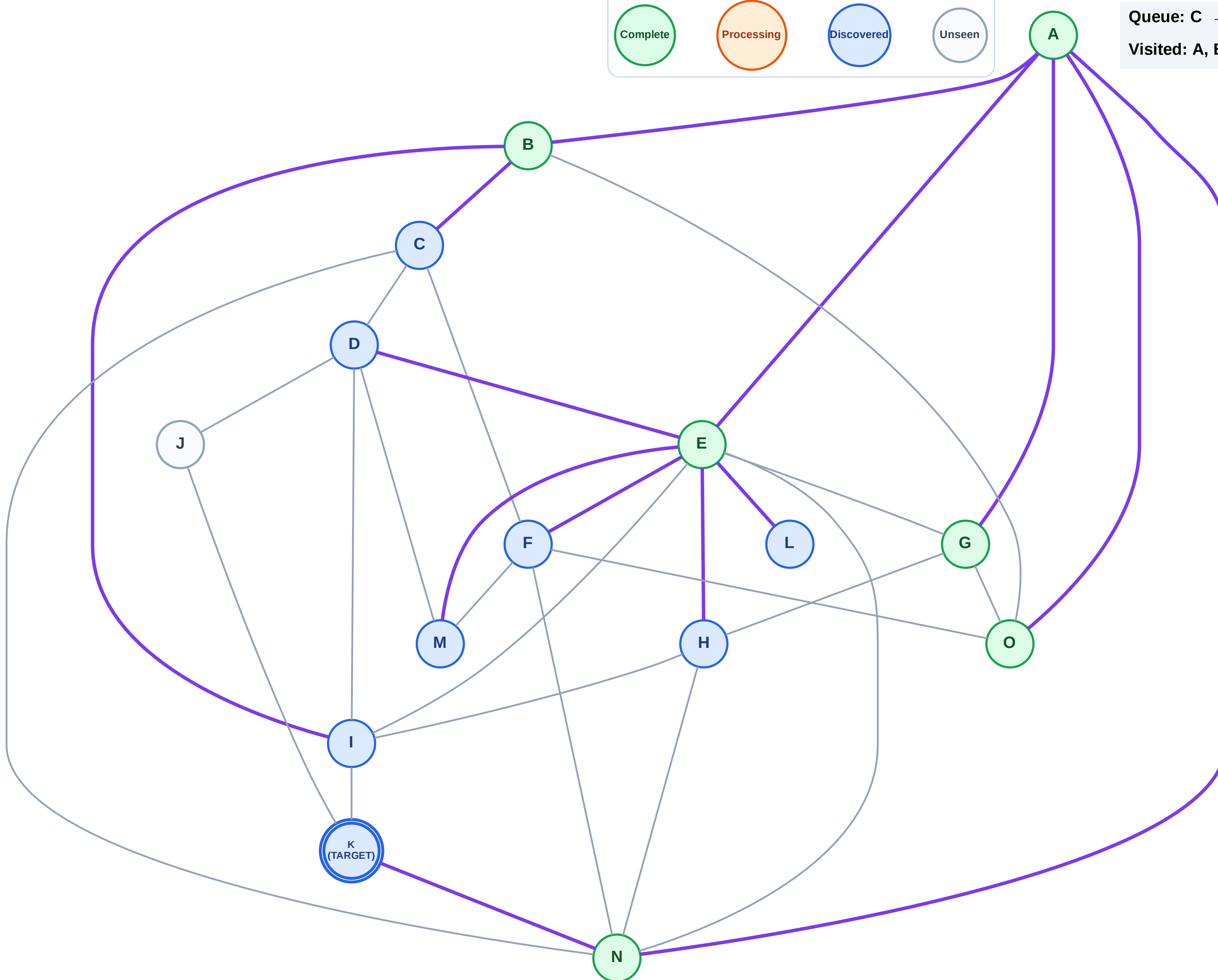


Discovered



Unseen

Queue: C → I → D → F → H → L → M → K
Visited: A, B, E, G, N, O



Step 14: Process node C

Legend

Legend

Complete

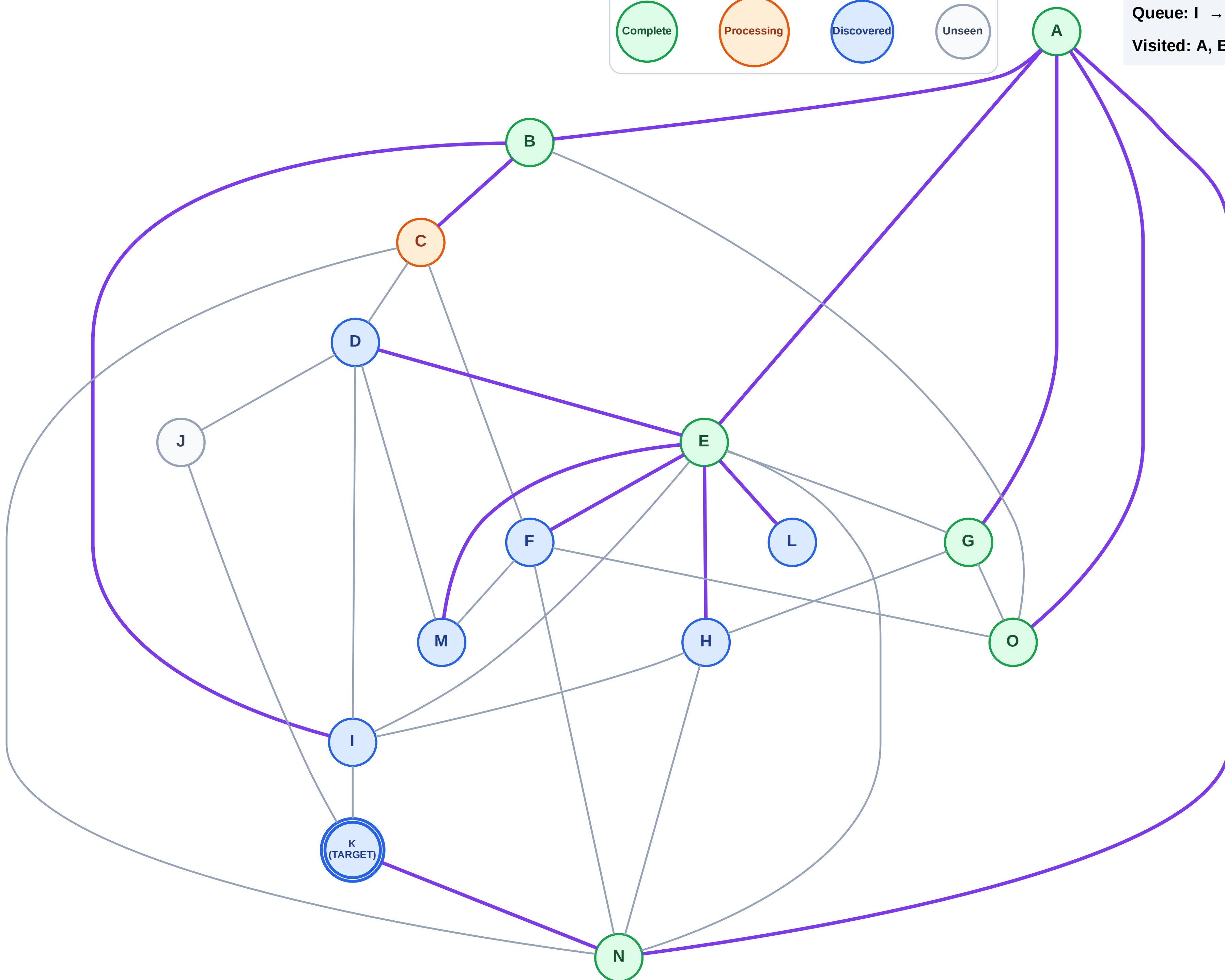
Processing

Discovered

Unseen

Queue: I → D → F → H → L → M → K

Visited: A, B, E, G, N, O



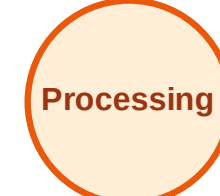
BFS Traversal

Step 15: No new neighbors from C

Legend



Complete



Processing

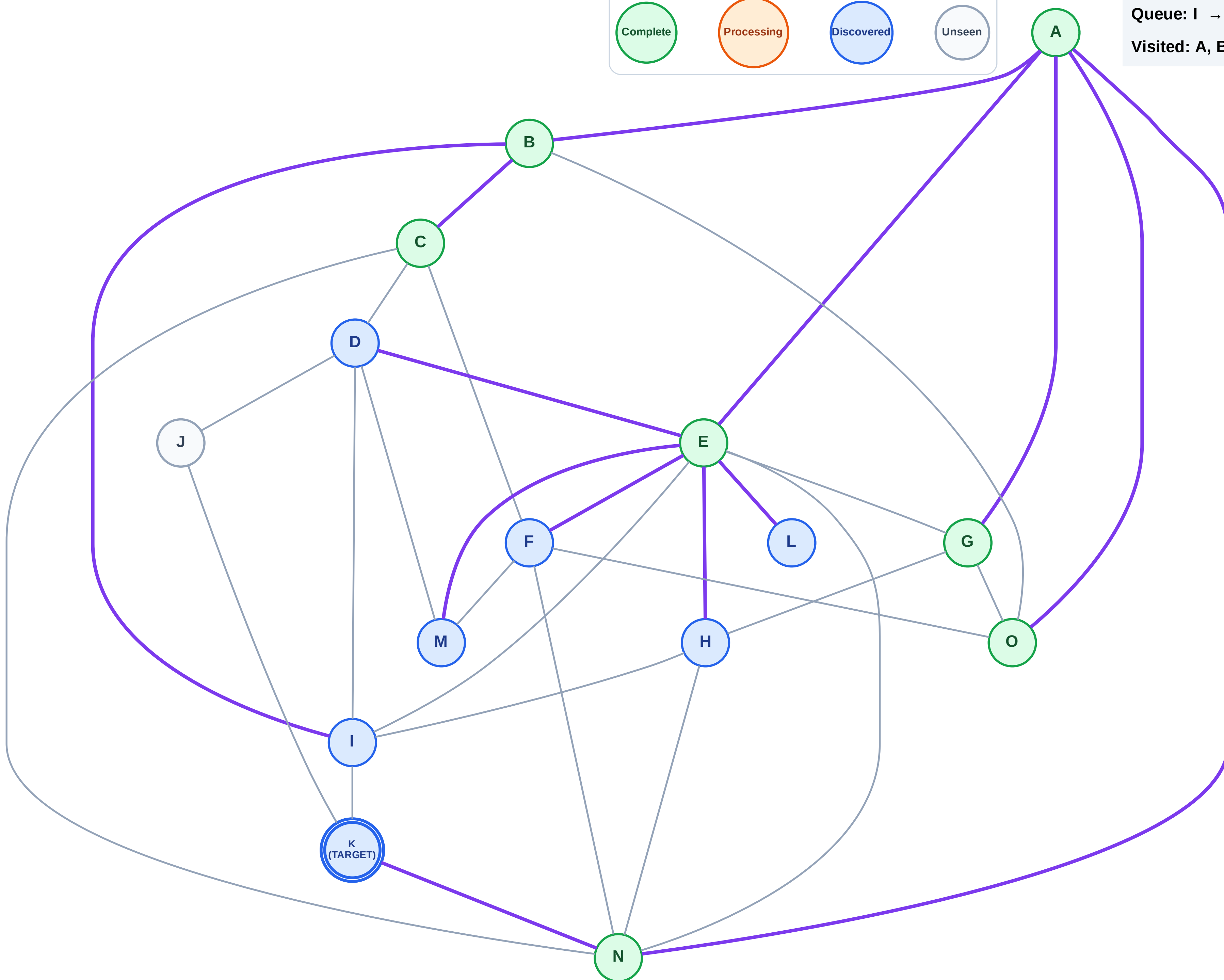


Discovered



Unseen

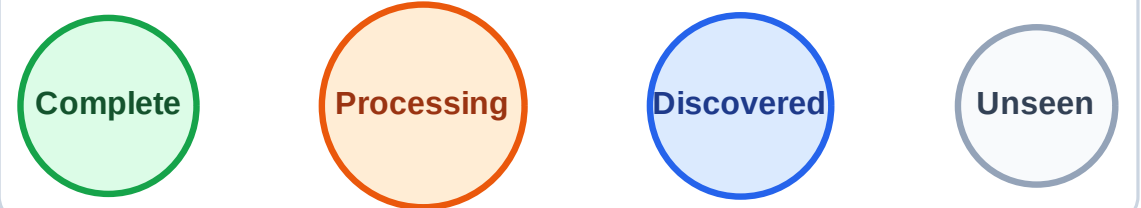
Queue: I → D → F → H → L → M → K
Visited: A, B, C, E, G, N, O



BFS Traversal

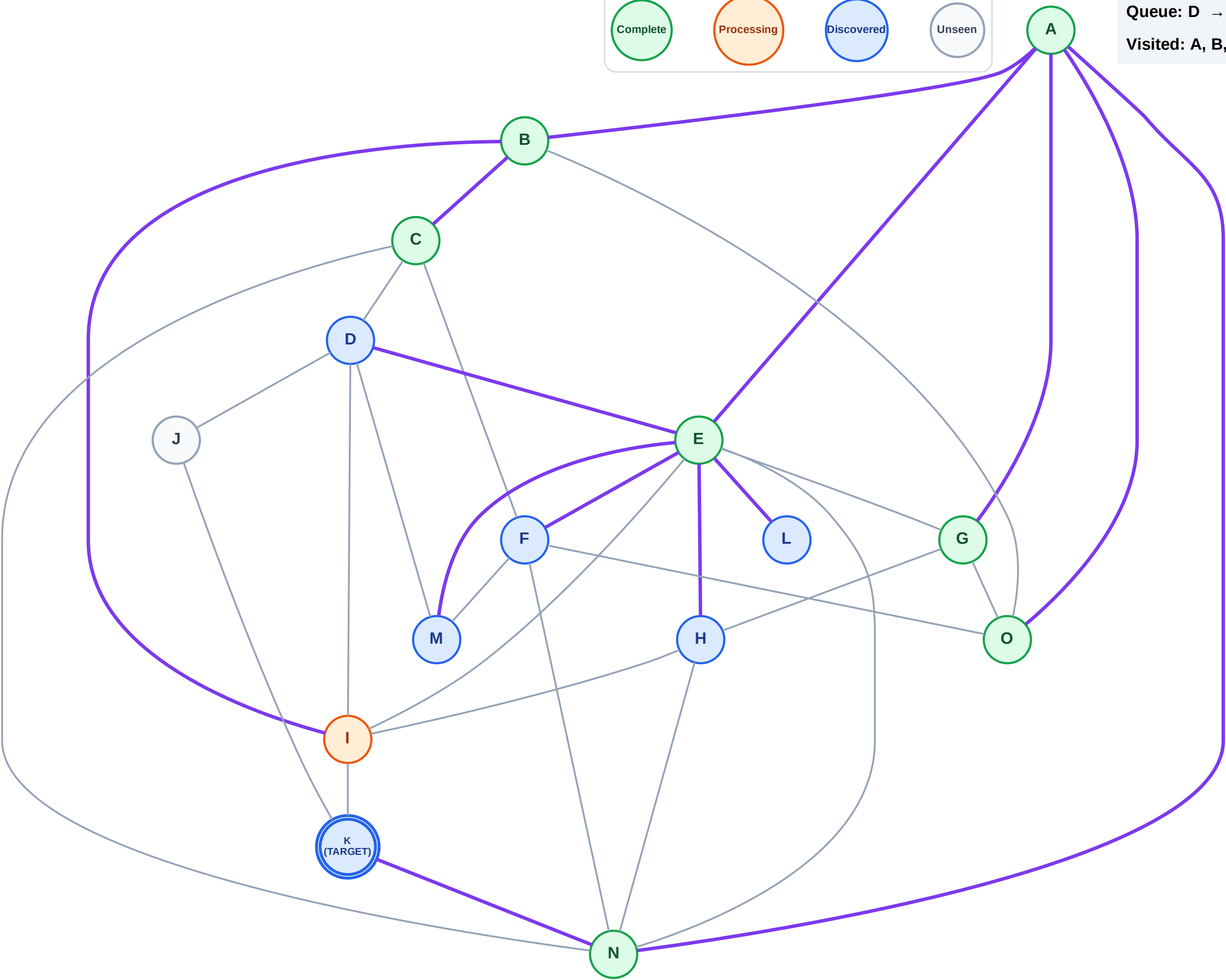
Step 16: Process node I

Legend



Queue: D → F → H → L → M → K

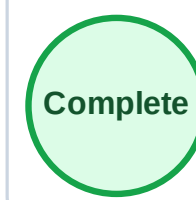
Visited: A, B, C, E, G, N, O



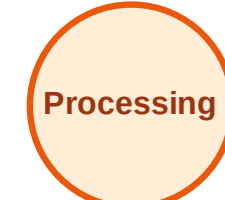
BFS Traversal

Step 17: No new neighbors from I

Legend



Complete



Processing



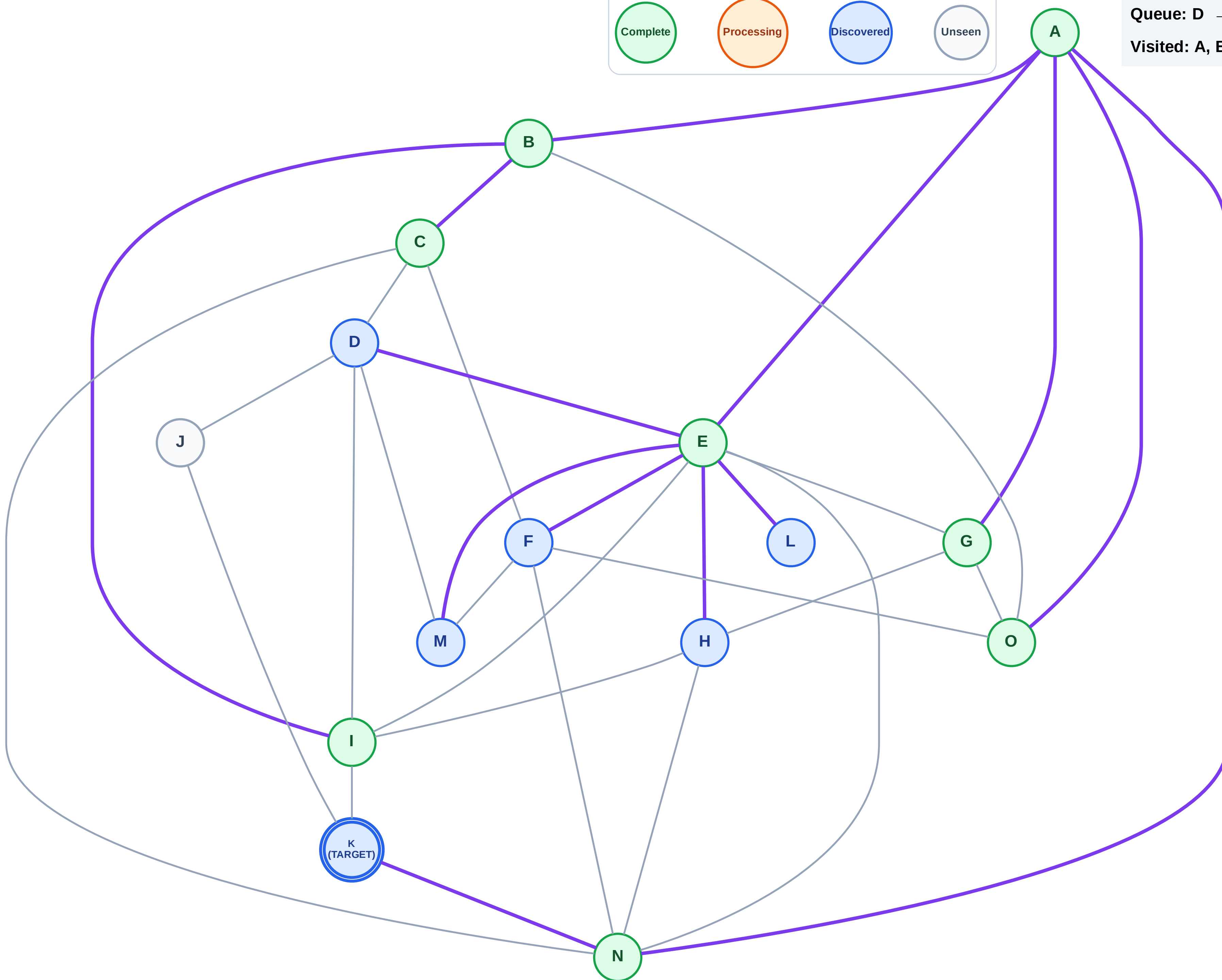
Discovered



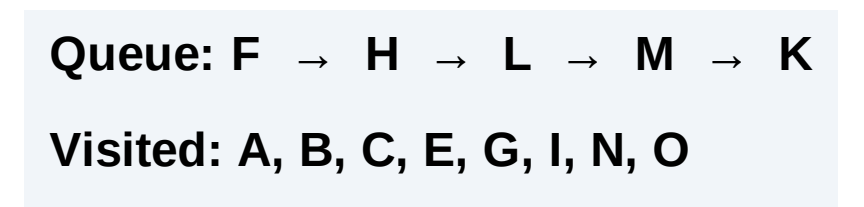
Unseen

Queue: D → F → H → L → M → K

Visited: A, B, C, E, G, I, N, O



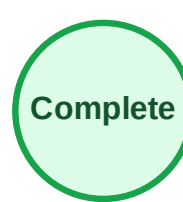
Step 18: Process node D



BFS Traversal

Step 19: Discover J from D

Legend



Complete



Processing

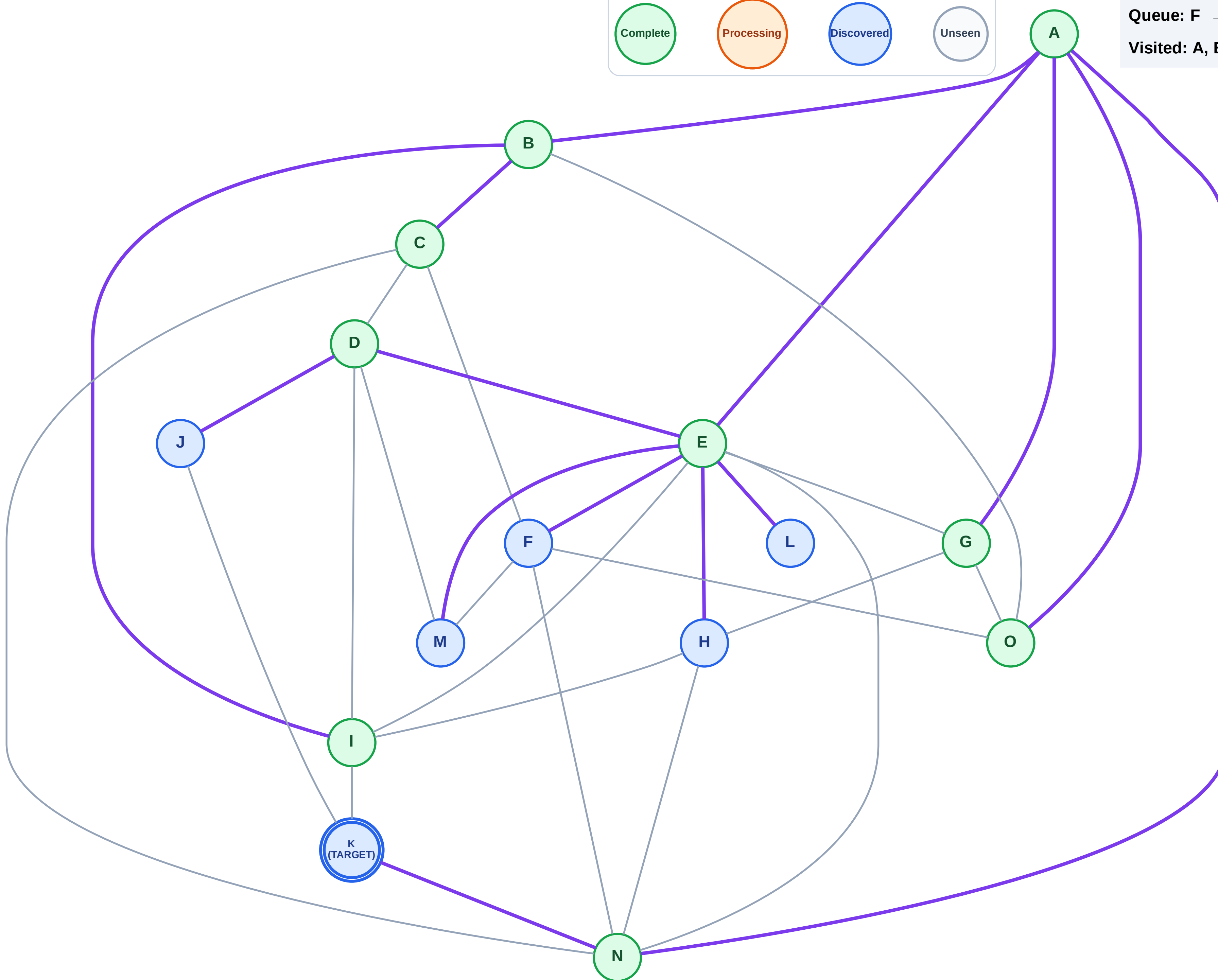


Discovered



Unseen

Queue: F → H → L → M → K → J
Visited: A, B, C, D, E, G, I, N, O



BFS Traversal

Step 20: Process node F

Legend

Complete

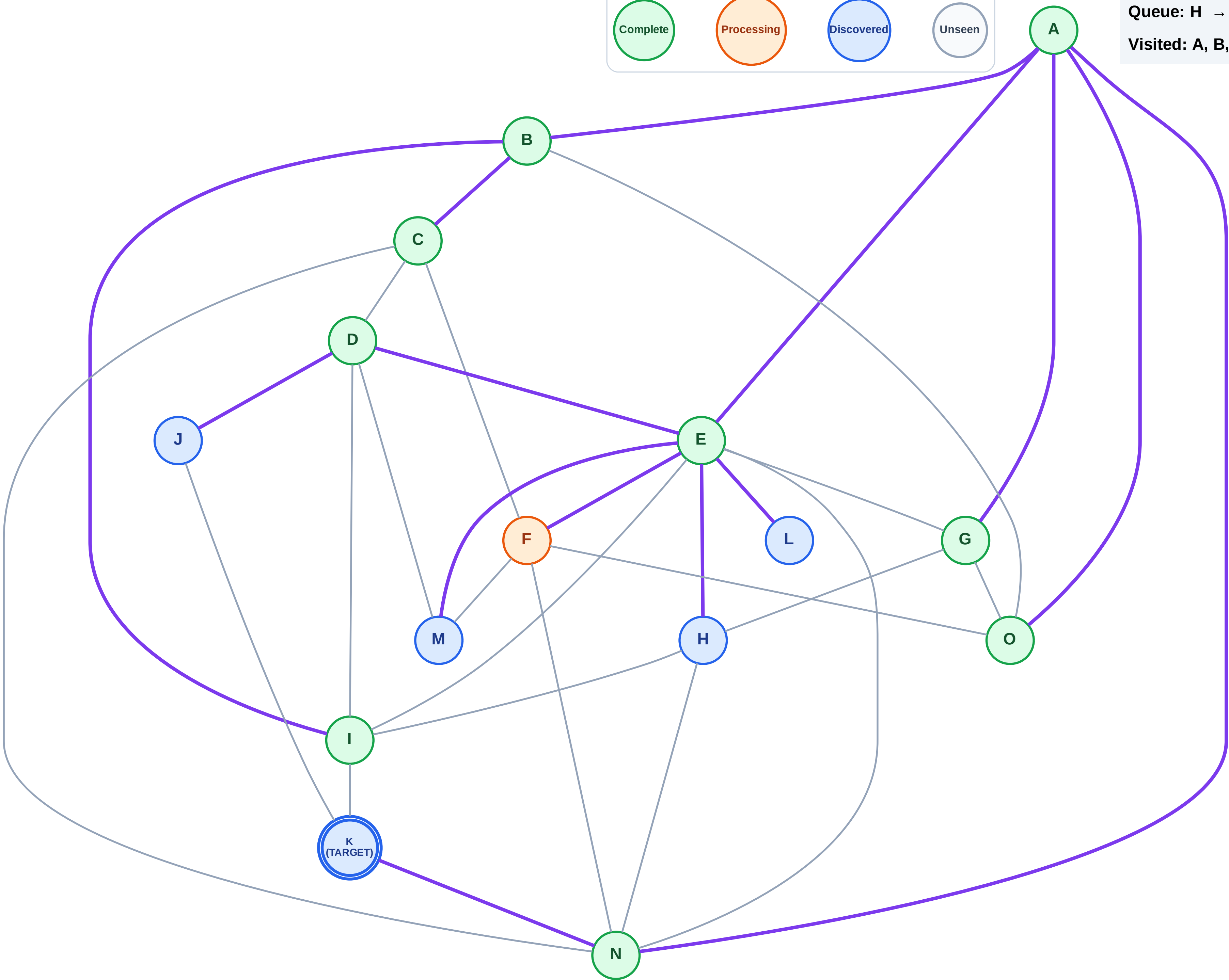
Processing

Discovered

Unseen

Queue: H → L → M → K → J

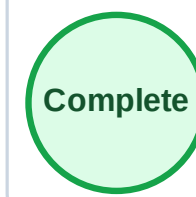
Visited: A, B, C, D, E, G, I, N, O



BFS Traversal

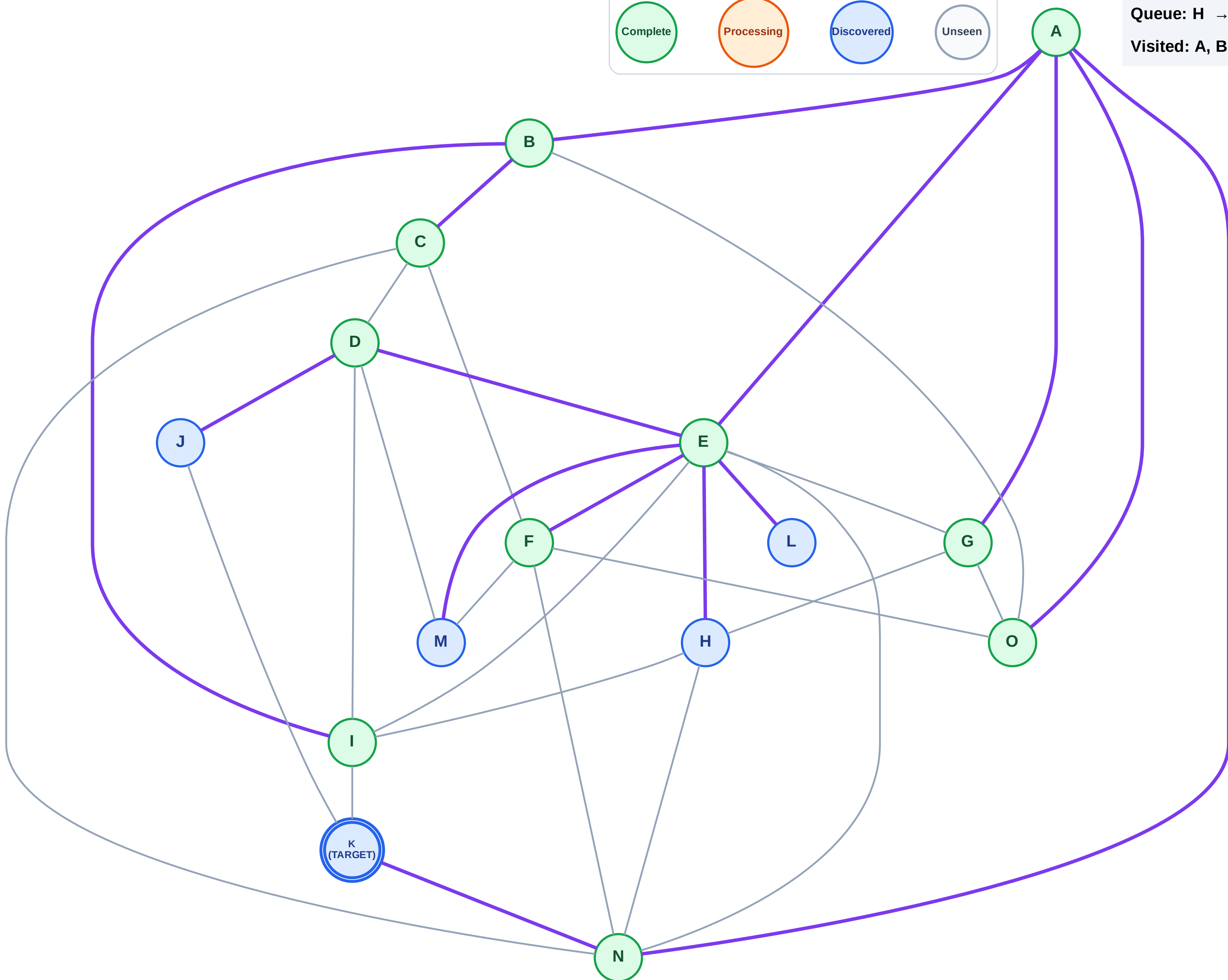
Step 21: No new neighbors from F

Legend



Queue: H → L → M → K → J

Visited: A, B, C, D, E, F, G, I, N, O



BFS Traversal

Step 22: Process node H

Legend

Complete

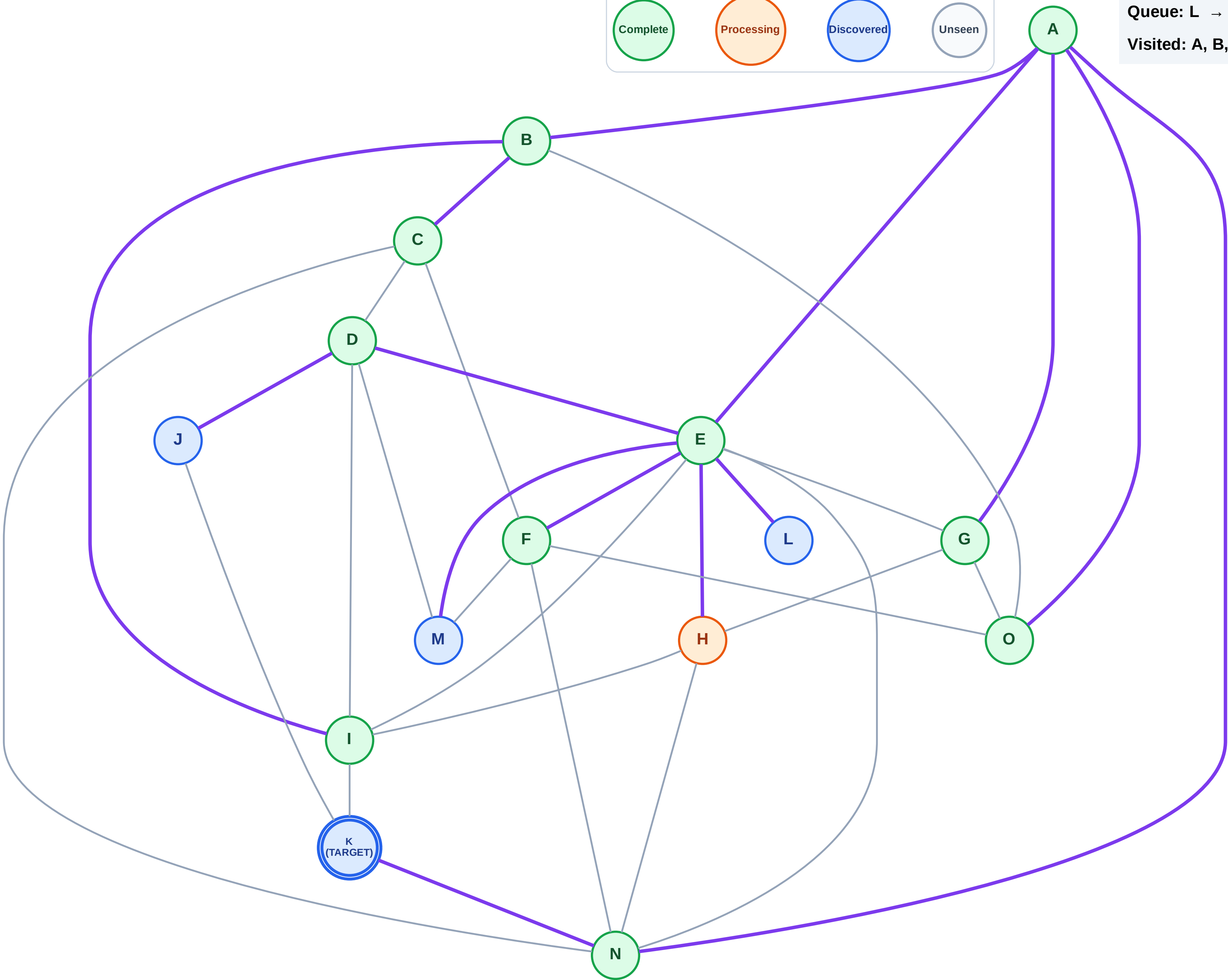
Processing

Discovered

Unseen

Queue: L → M → K → J

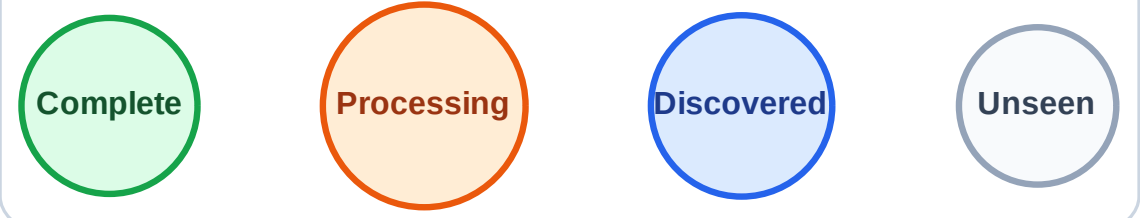
Visited: A, B, C, D, E, F, G, I, N, O



BFS Traversal

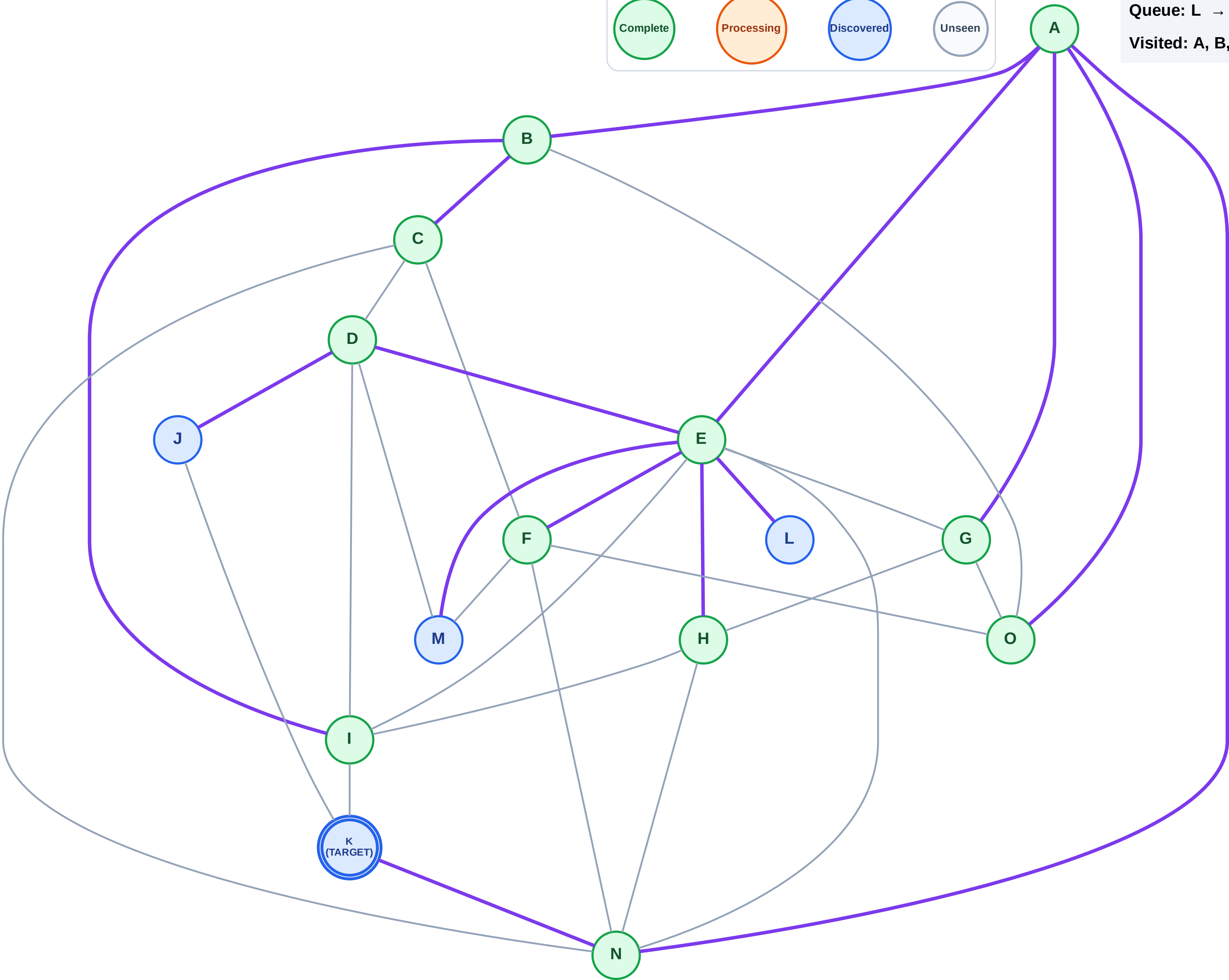
Step 23: No new neighbors from H

Legend



Queue: L → M → K → J

Visited: A, B, C, D, E, F, G, H, I, N, O



BFS Traversal

Step 24: Process node L

Legend



Complete



Processing



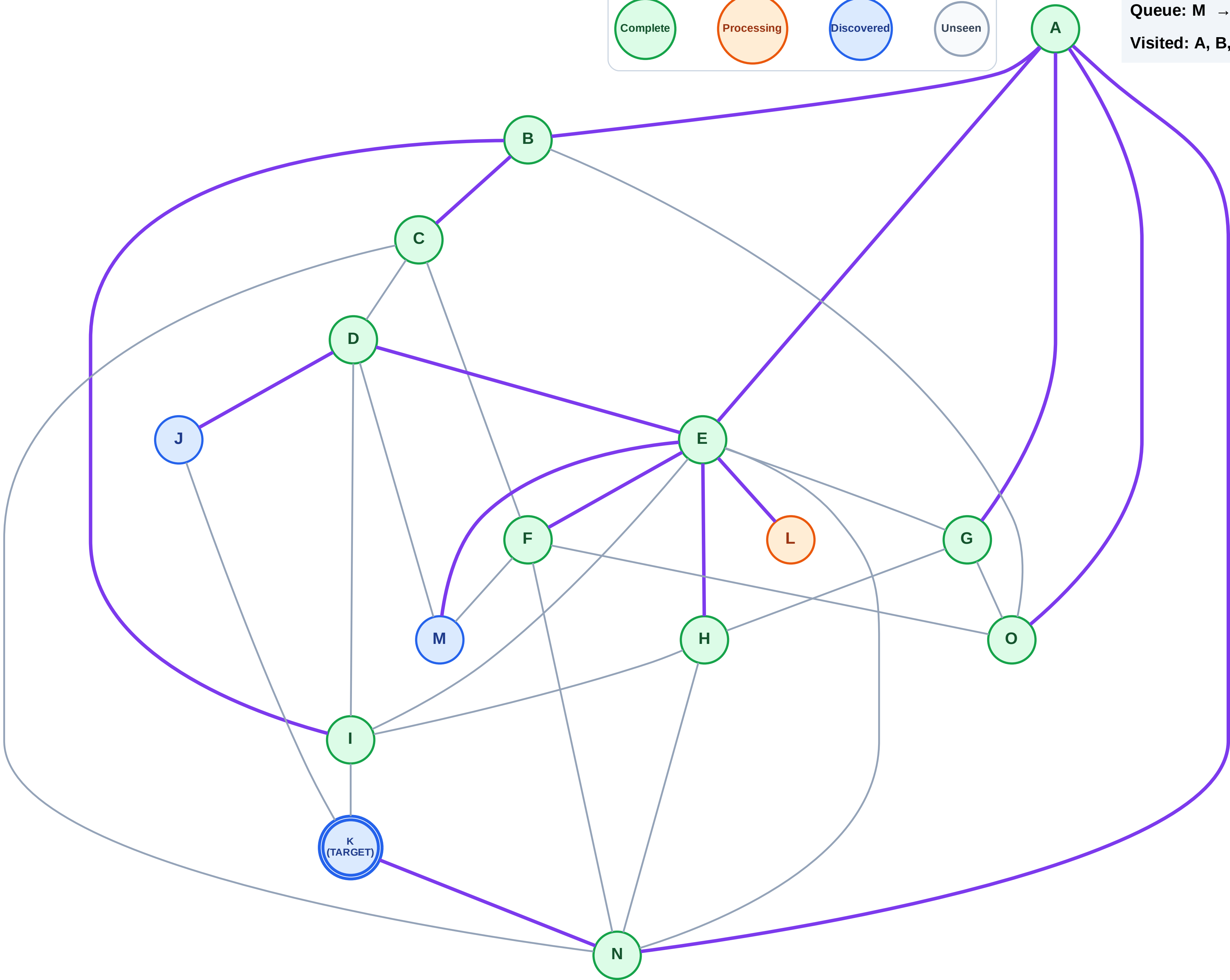
Discovered



Unseen

Queue: M → K → J

Visited: A, B, C, D, E, F, G, H, I, N, O



BFS Traversal

Step 25: No new neighbors from L

Legend



Complete



Processing



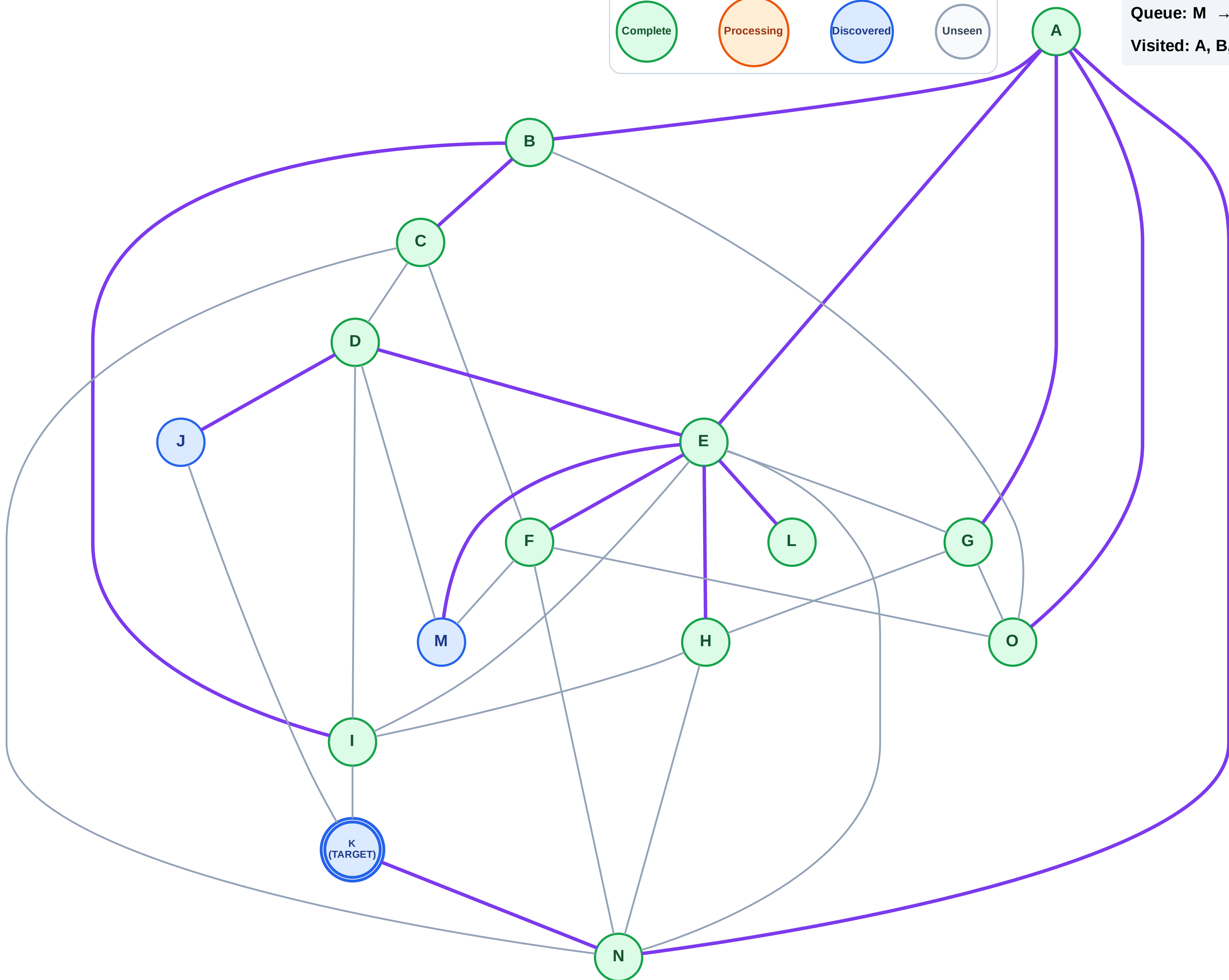
Discovered



Unseen

Queue: M → K → J

Visited: A, B, C, D, E, F, G, H, I, L, N, O



BFS Traversal

Step 26: Process node M

Legend

Complete

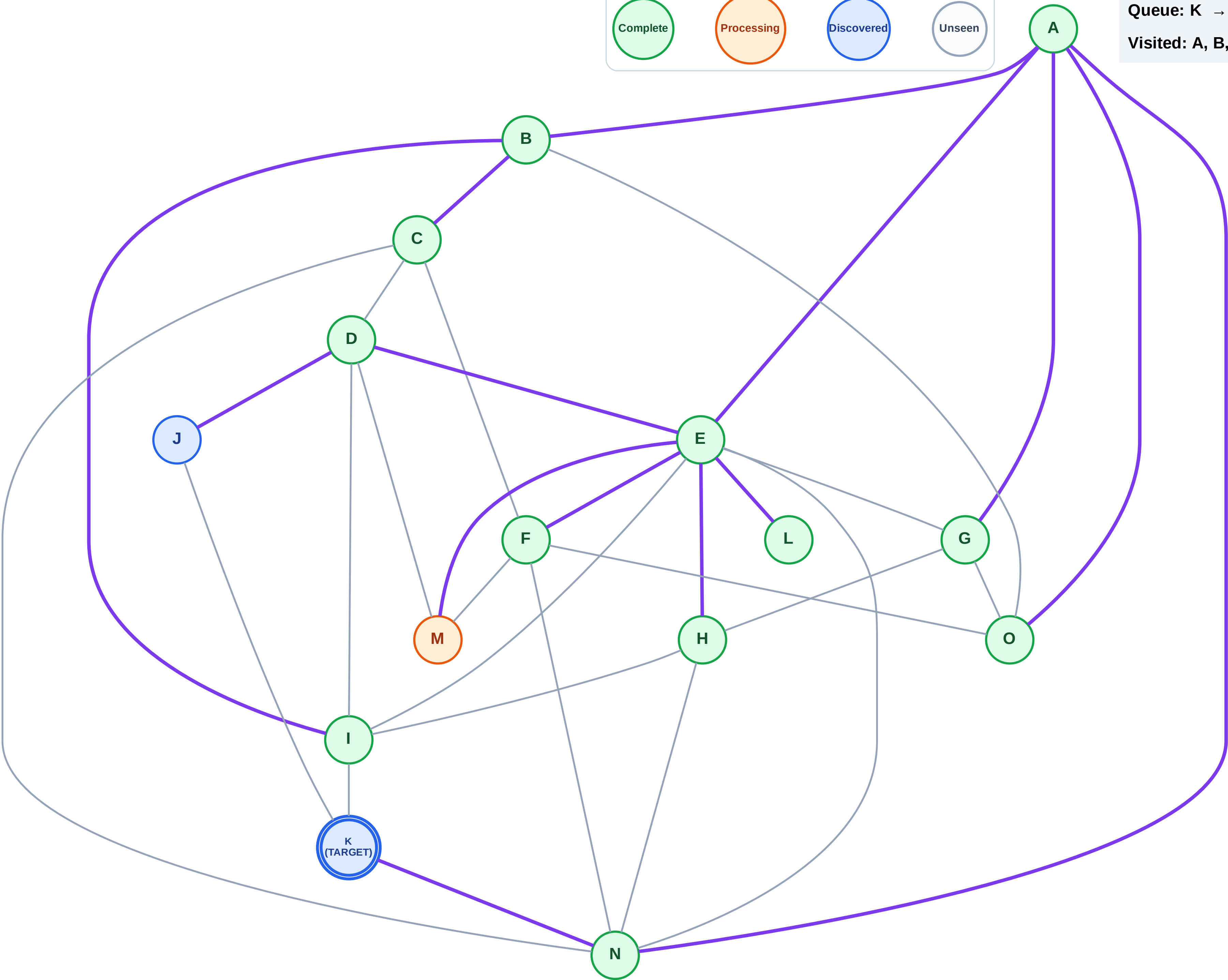
Processing

Discovered

Unseen

Queue: K → J

Visited: A, B, C, D, E, F, G, H, I, L, N, O



BFS Traversal

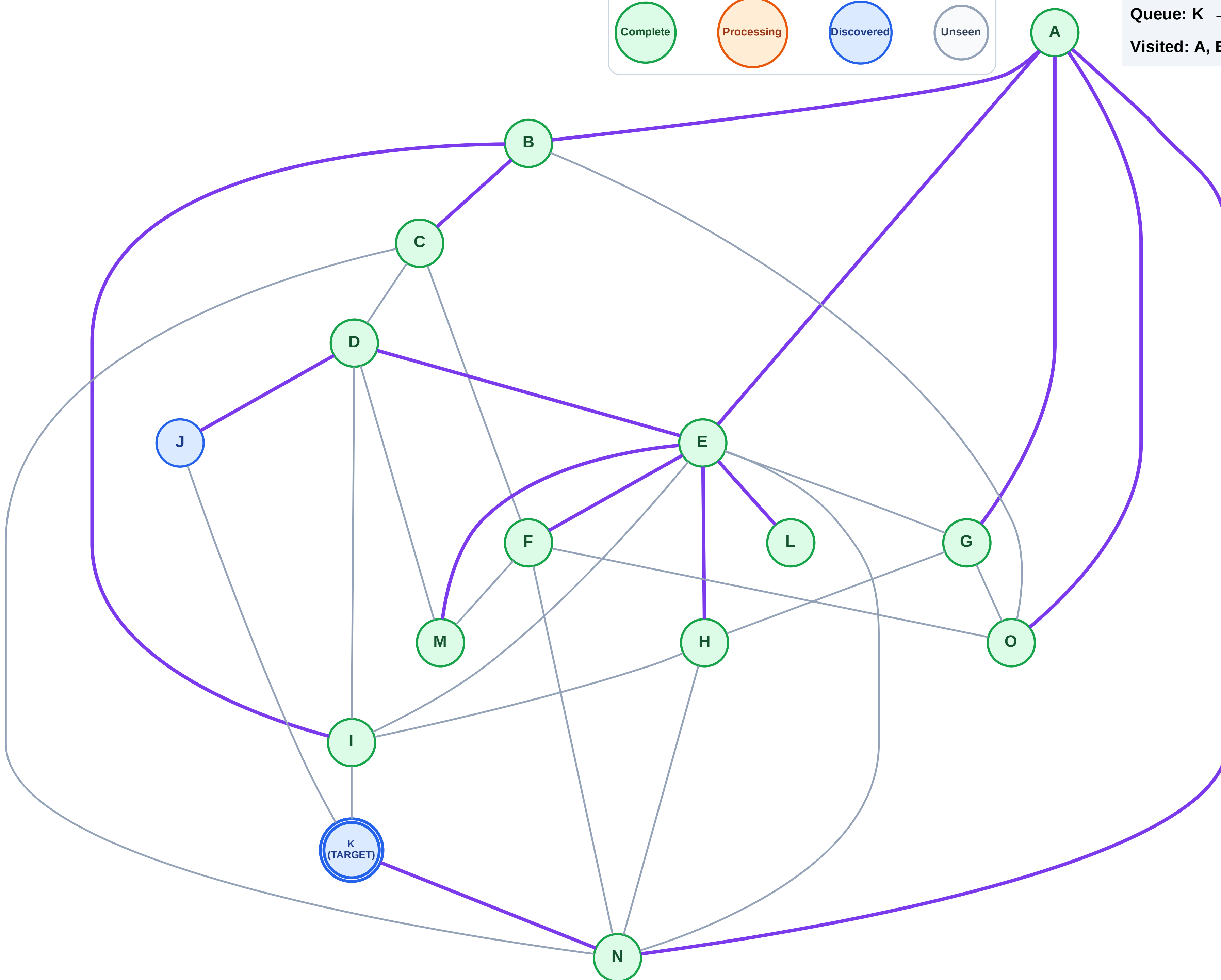
Step 27: No new neighbors from M

Legend



Queue: K → J

Visited: A, B, C, D, E, F, G, H, I, L, M, N, O



BFS Traversal

Step 28: Process node K

Legend



Complete



Processing



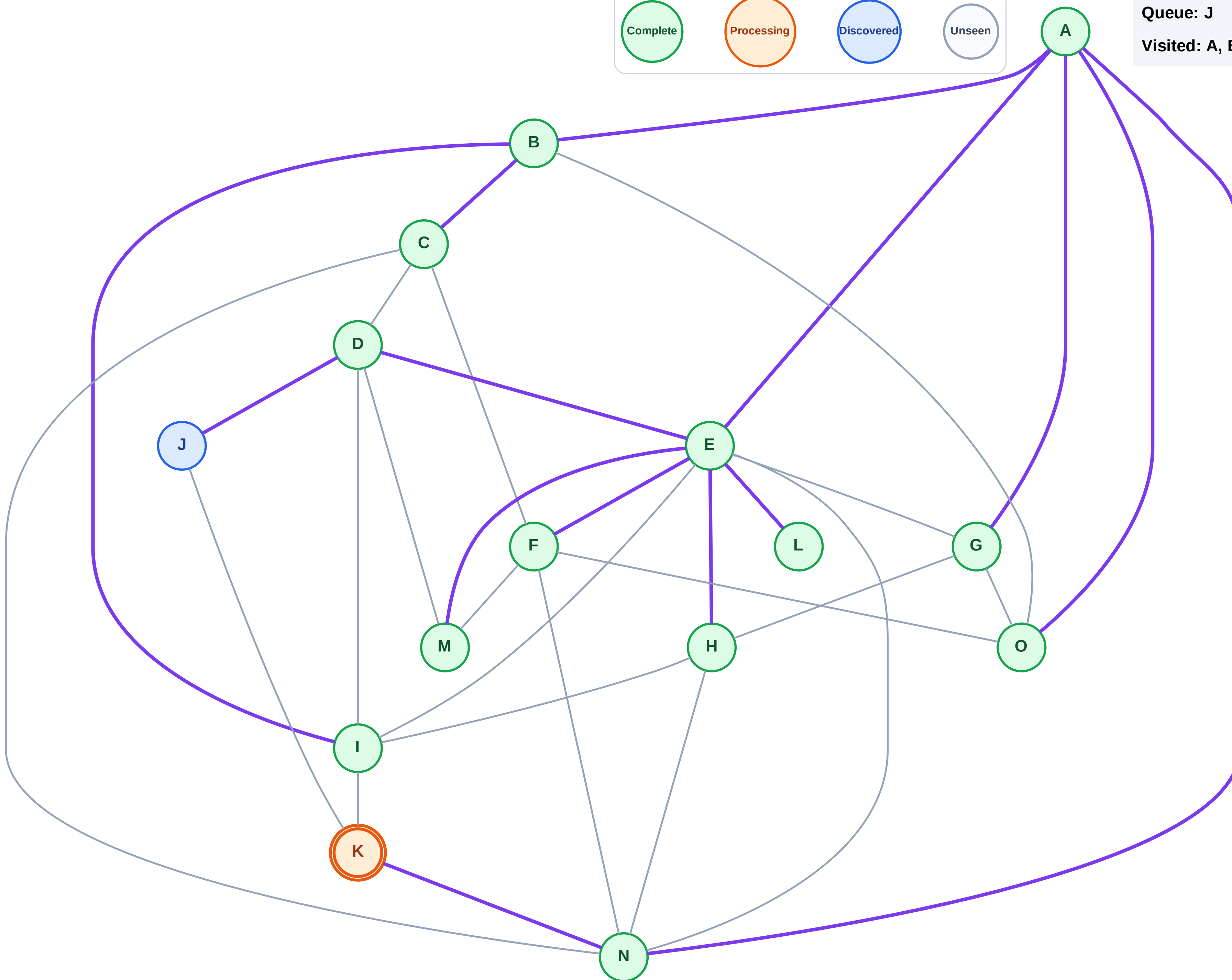
Discovered



Unseen

Queue: J

Visited: A, B, C, D, E, F, G, H, I, L, M, N, O



BFS Traversal

Step 29: TARGET NODE K REACHED! Path (2 hops): A → N → K

Legend



Complete



Processing



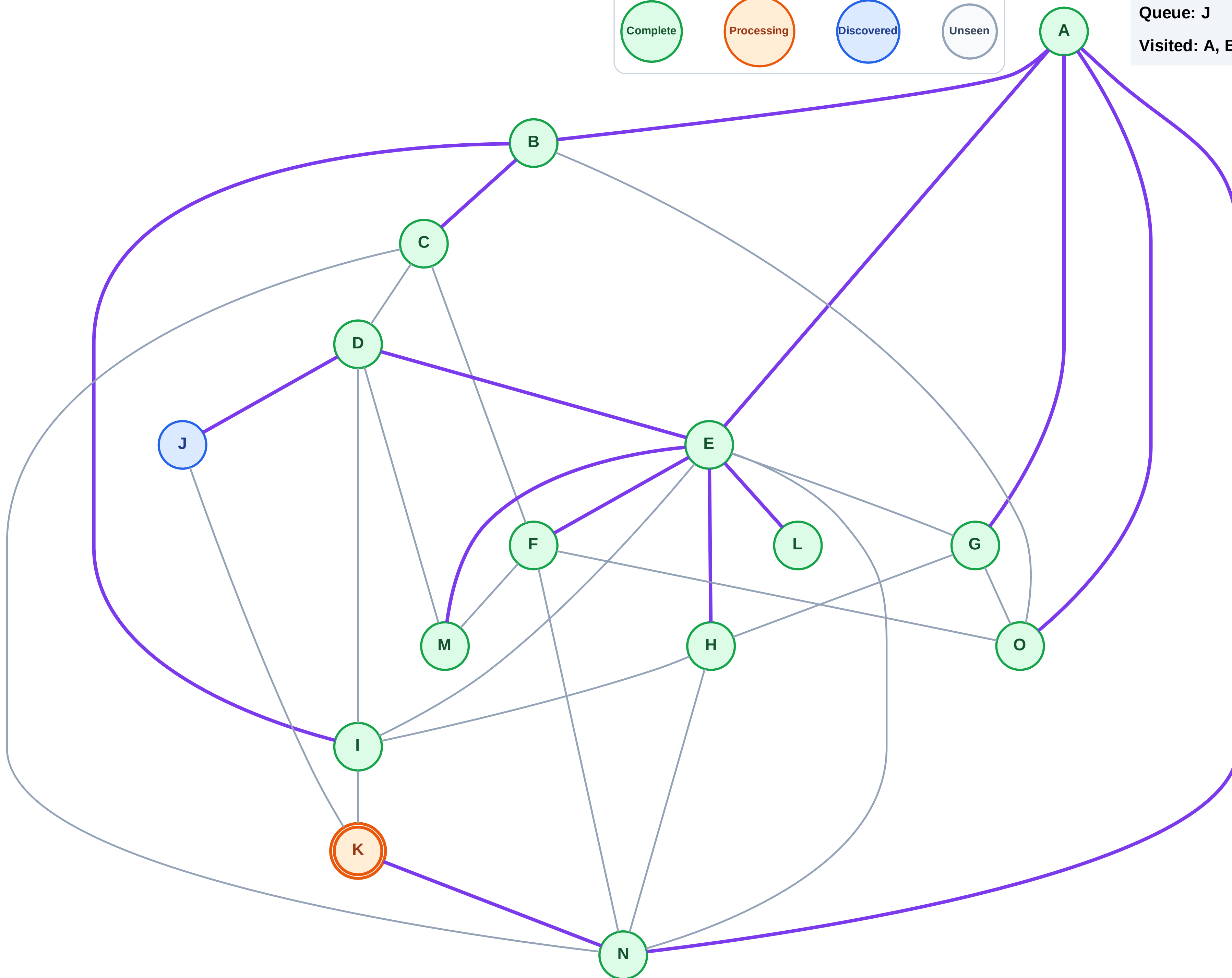
Discovered



Unseen

Queue: J

Visited: A, B, C, D, E, F, G, H, I, K, L, M, N, O



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

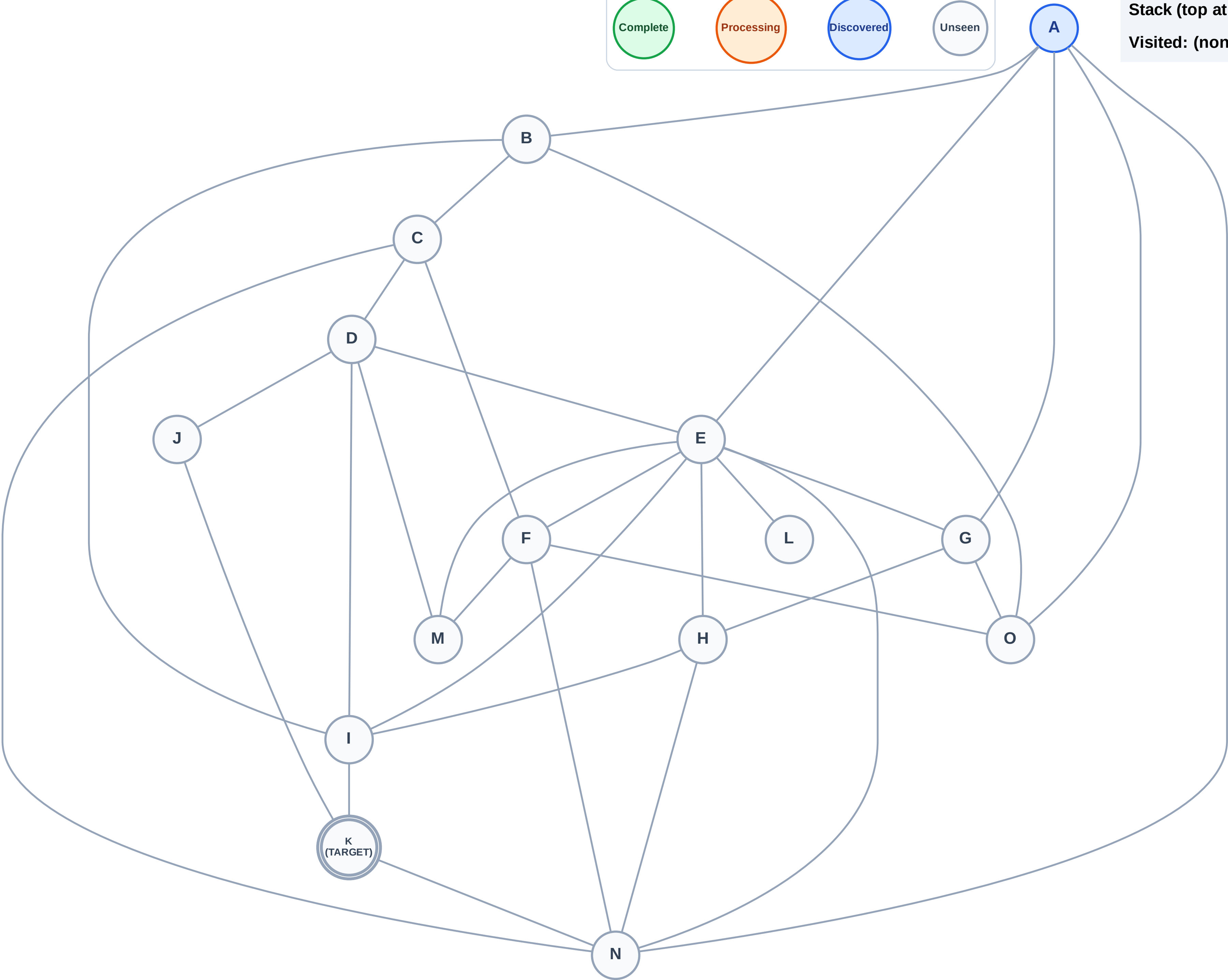
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

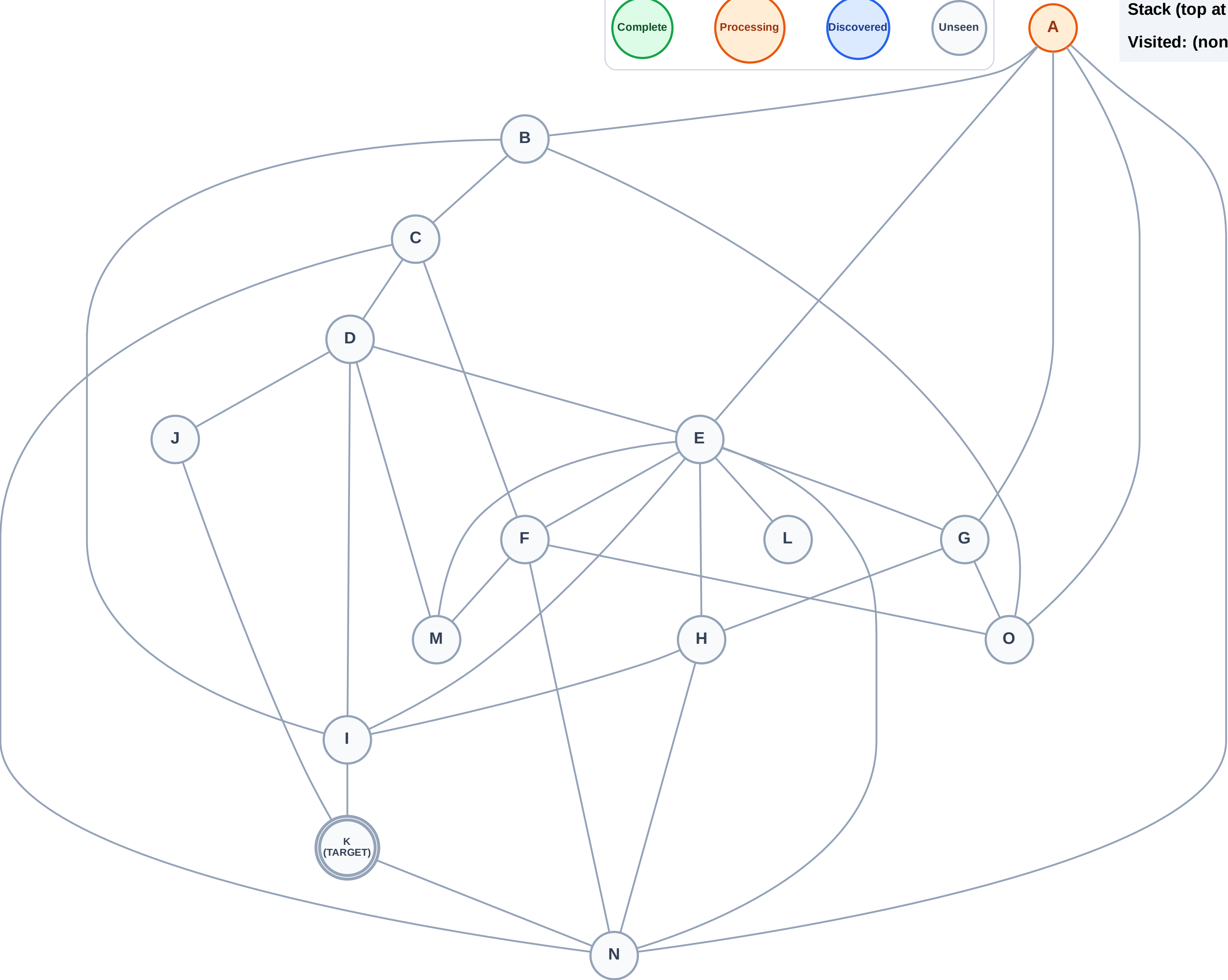
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

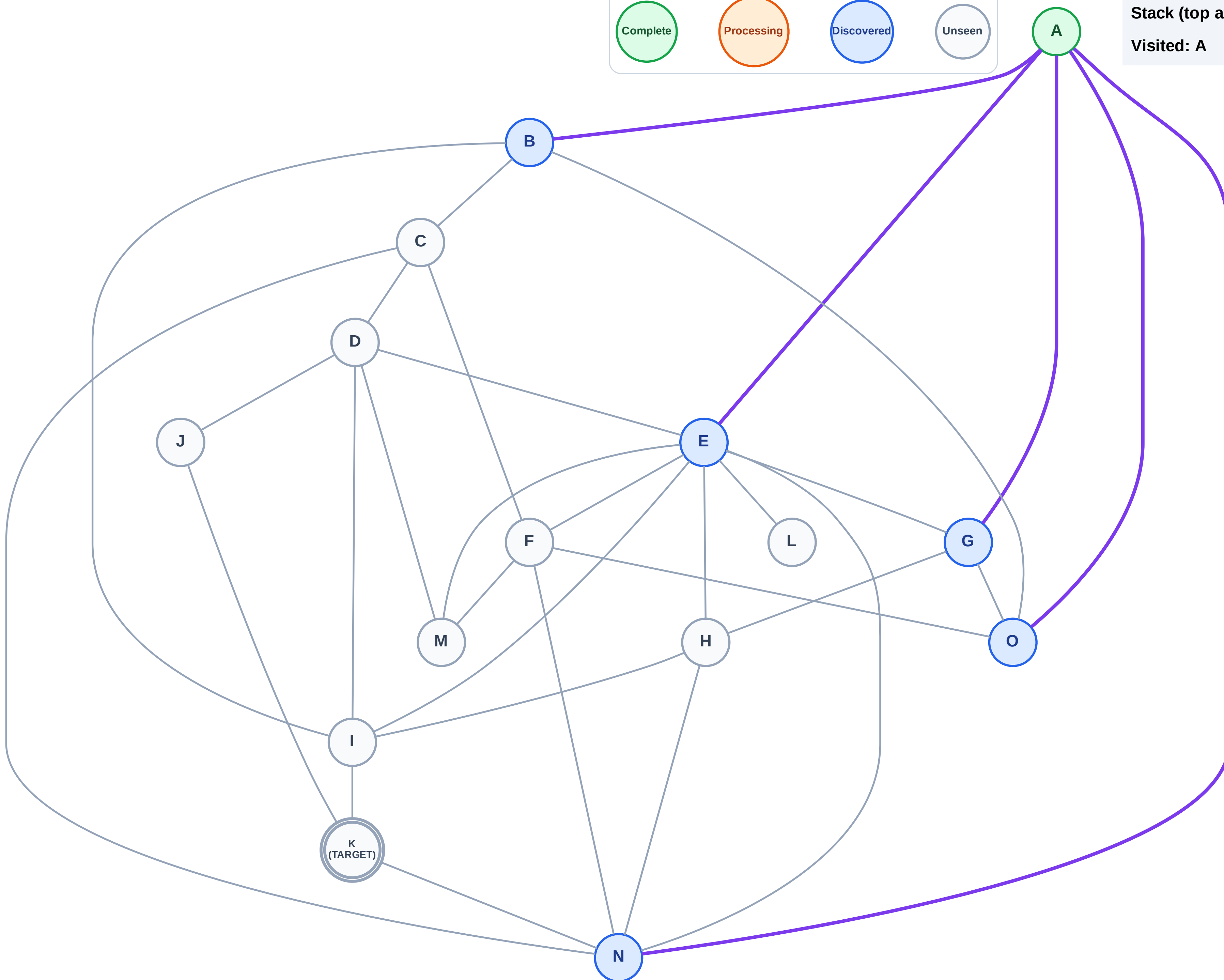
Step 3: Discover O, N, G, E, B from A

Legend



Stack (top at right): O | N | G | E | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

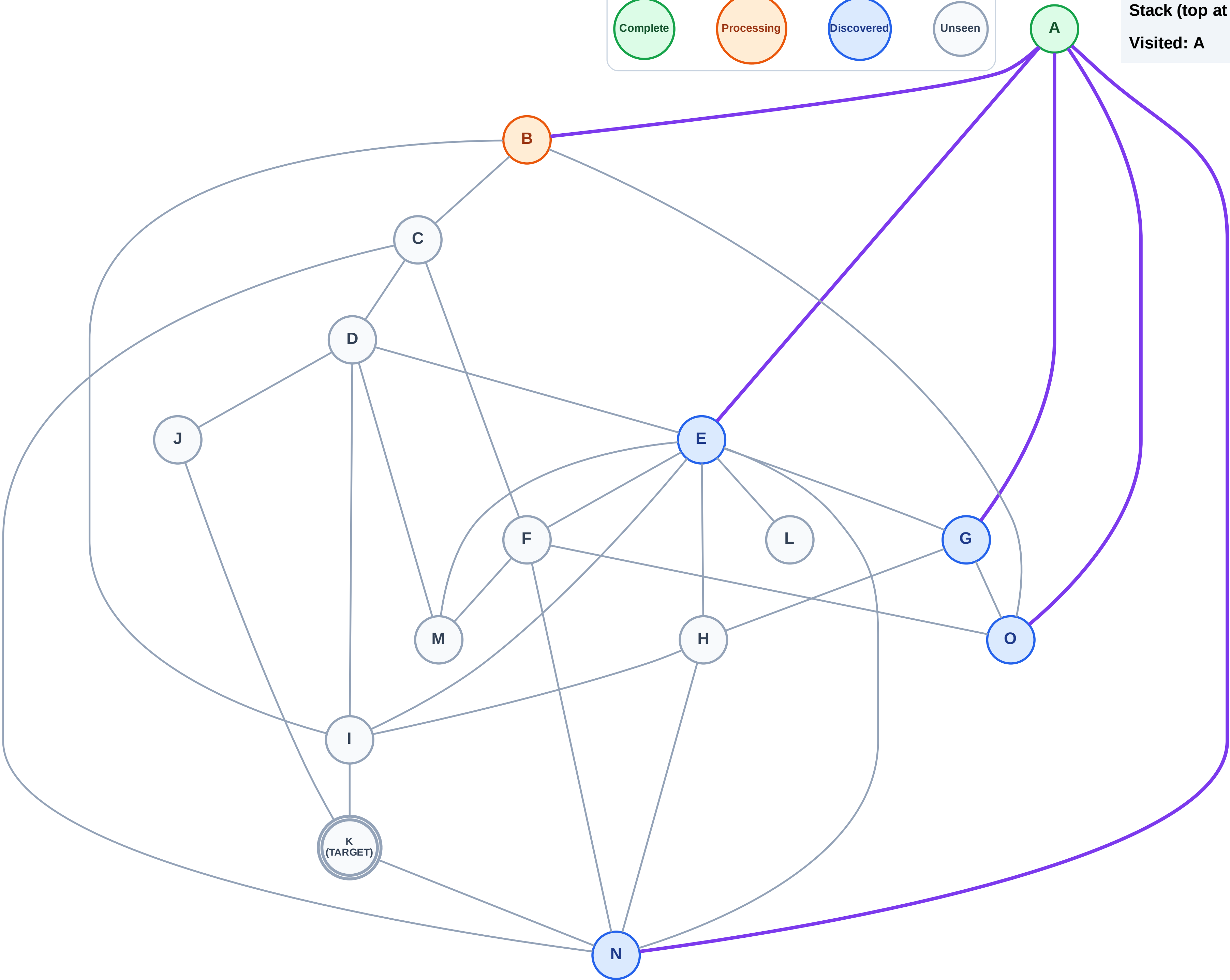
Processing

Discovered

Unseen

Stack (top at right): O | N | G | E

Visited: A



Step 5: Discover I, C from

Complete

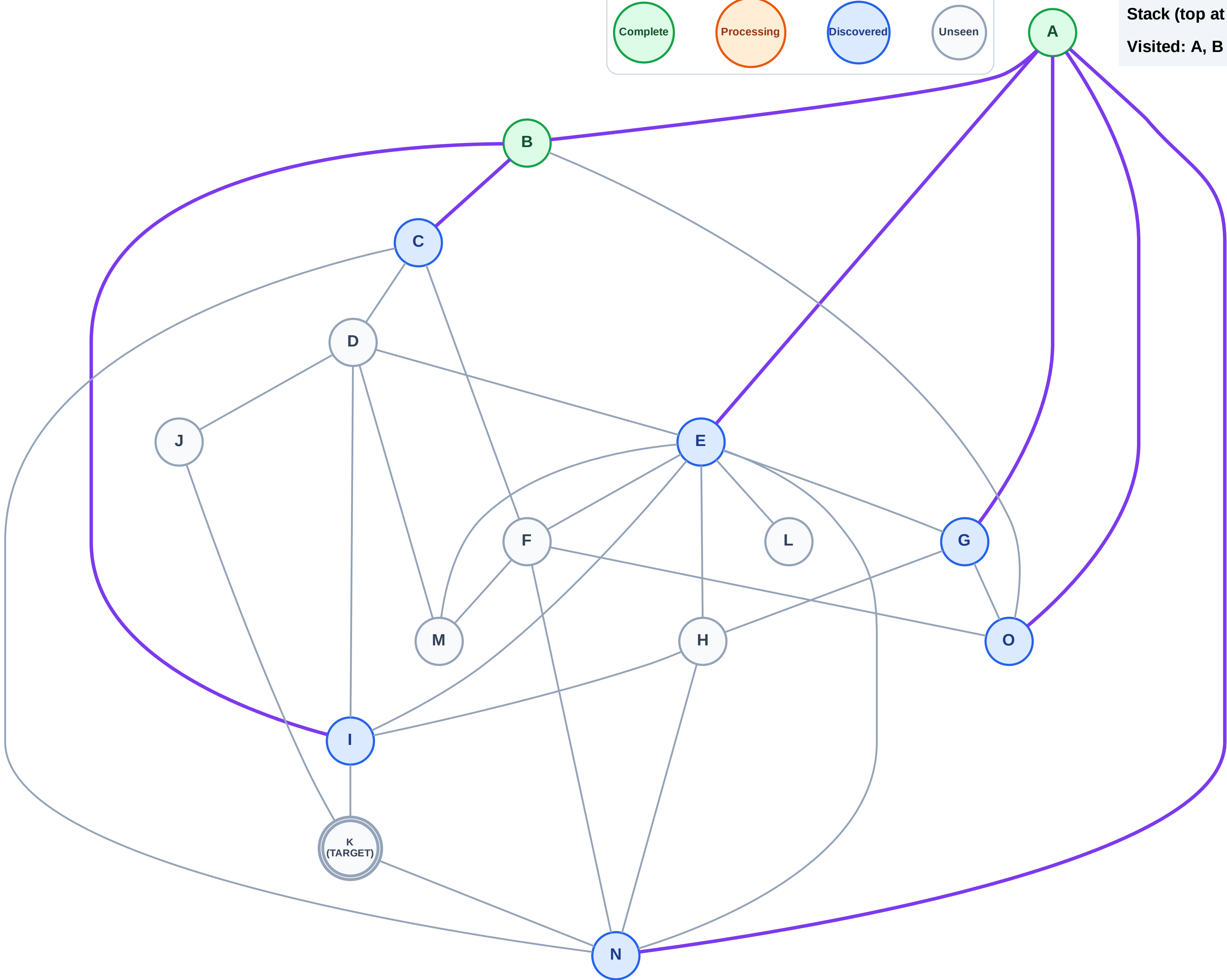


Processing

Discovered

Unseen

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend



Complete



Processing



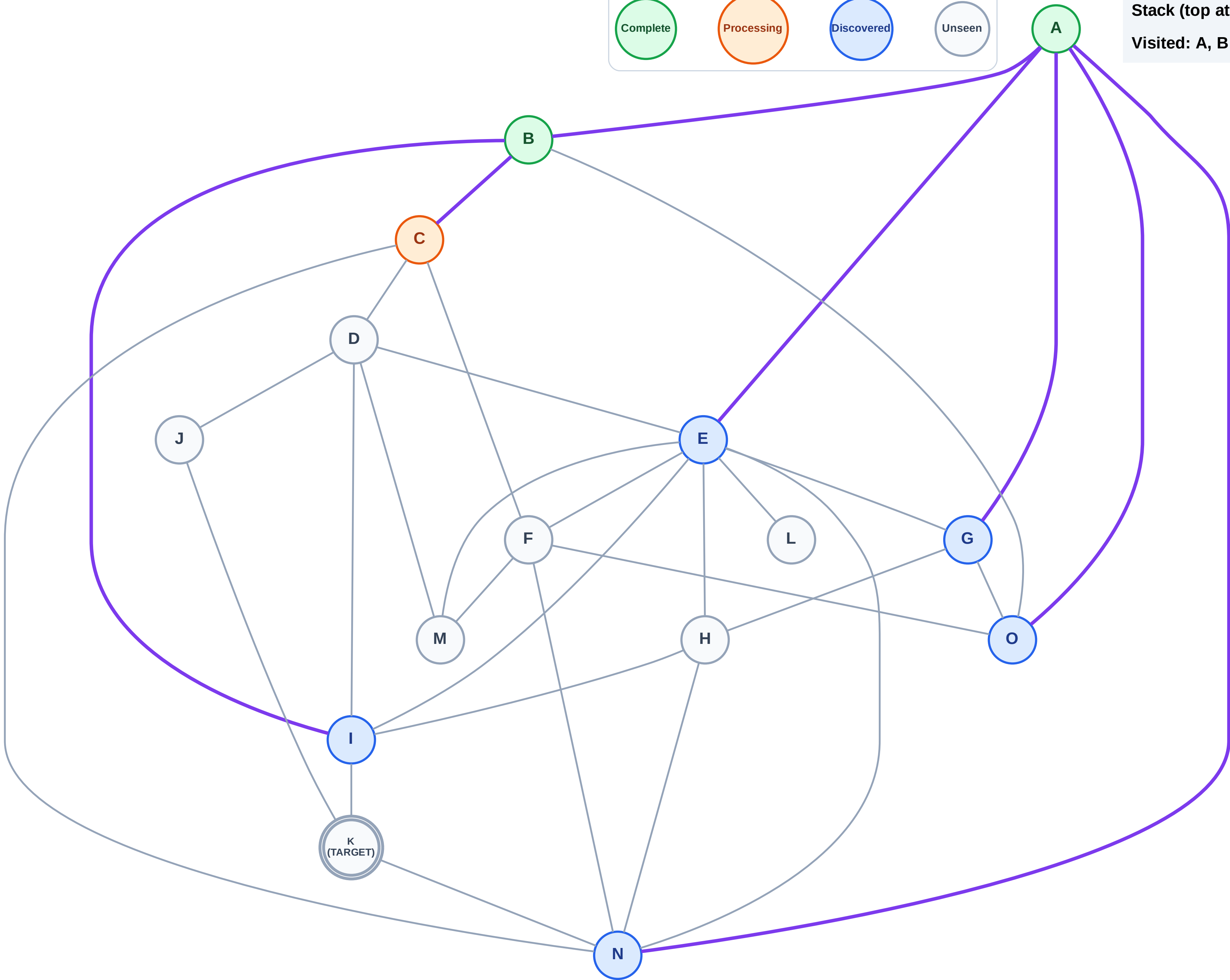
Discovered



Unseen

Stack (top at right): O | N | G | E | I

Visited: A, B



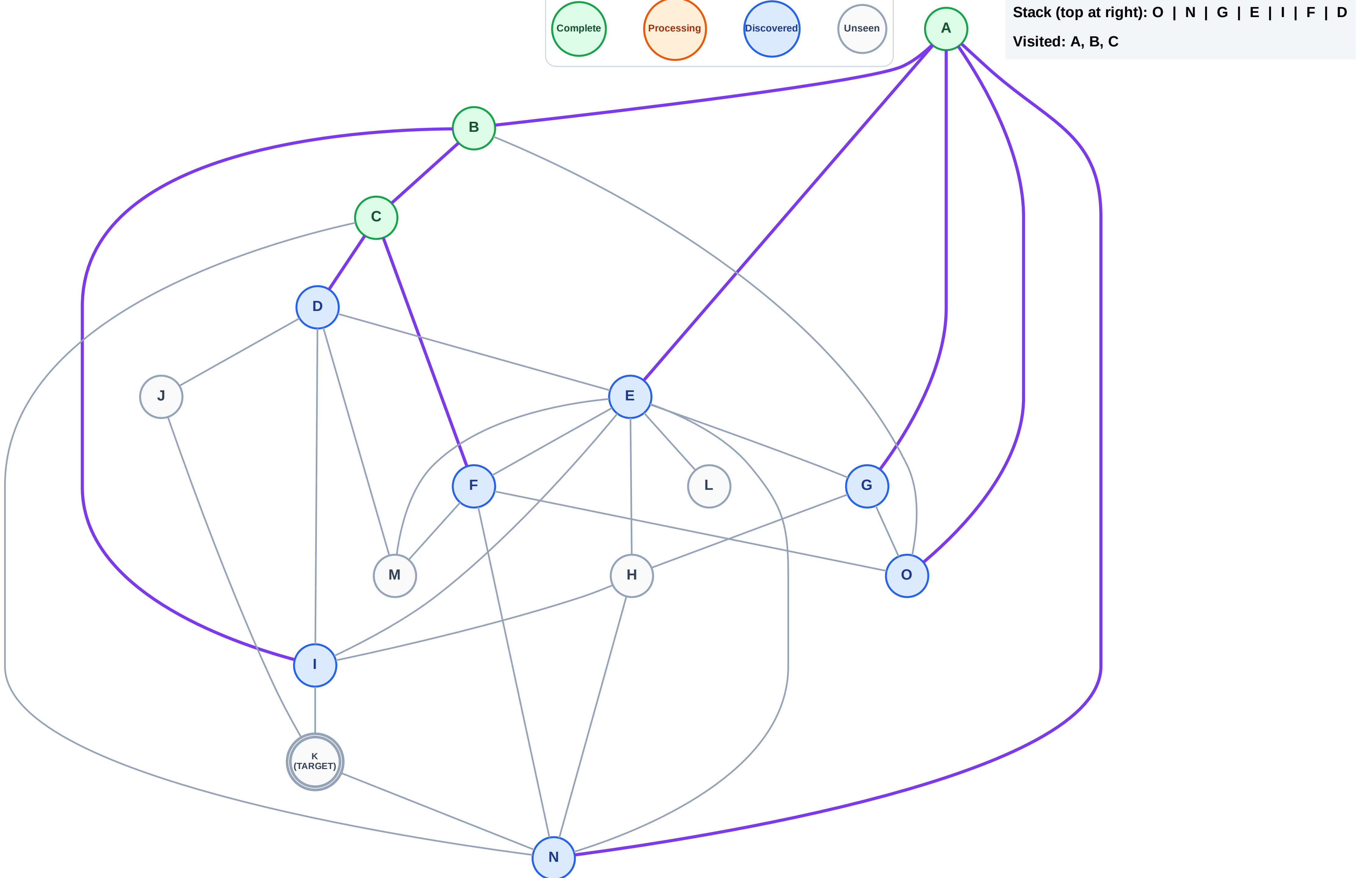
Step 7: Discover F, D from C

Legend

Complete Processing Discovered Unseen

Unseen

Visited: A, B, C



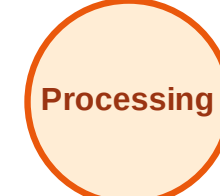
DFS Traversal

Step 8: Process node D

Legend



Complete



Processing



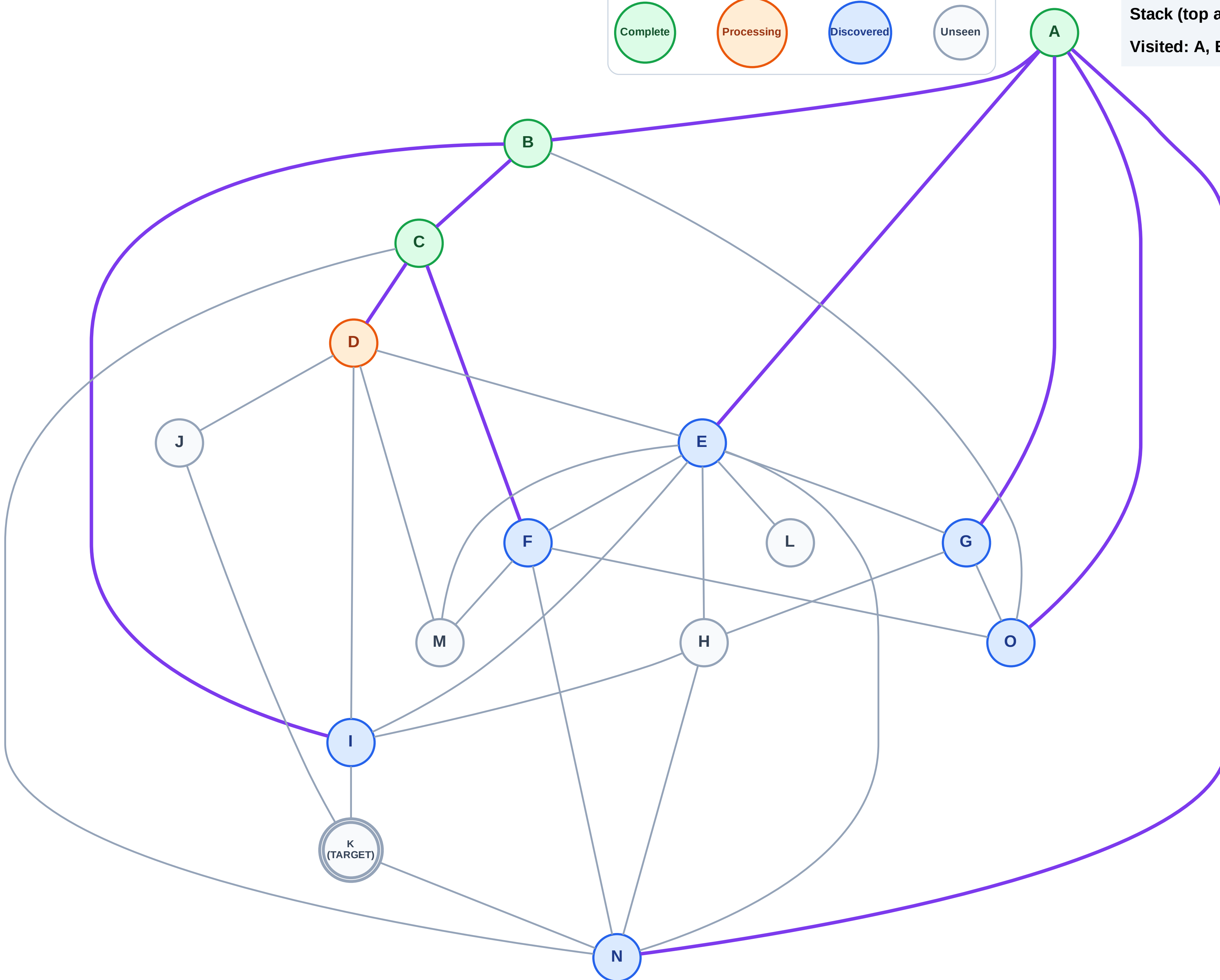
Discovered



Unseen

Stack (top at right): O | N | G | E | I | F

Visited: A, B, C



DFS Traversal

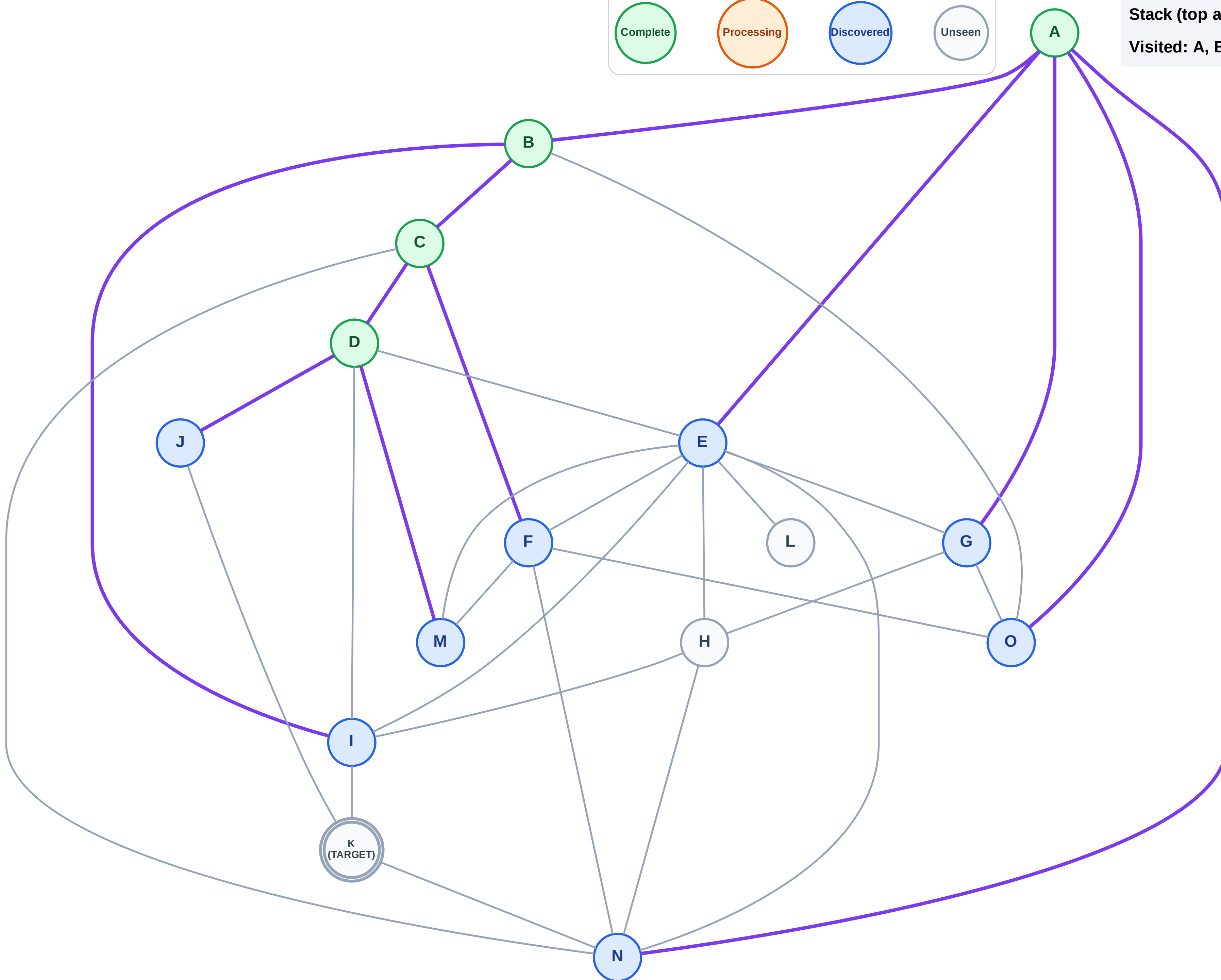
Step 9: Discover M, J from D

Legend



Stack (top at right): O | N | G | E | I | F | M | J

Visited: A, B, C, D



DFS Traversal

Step 10: Process node J

Legend

Complete

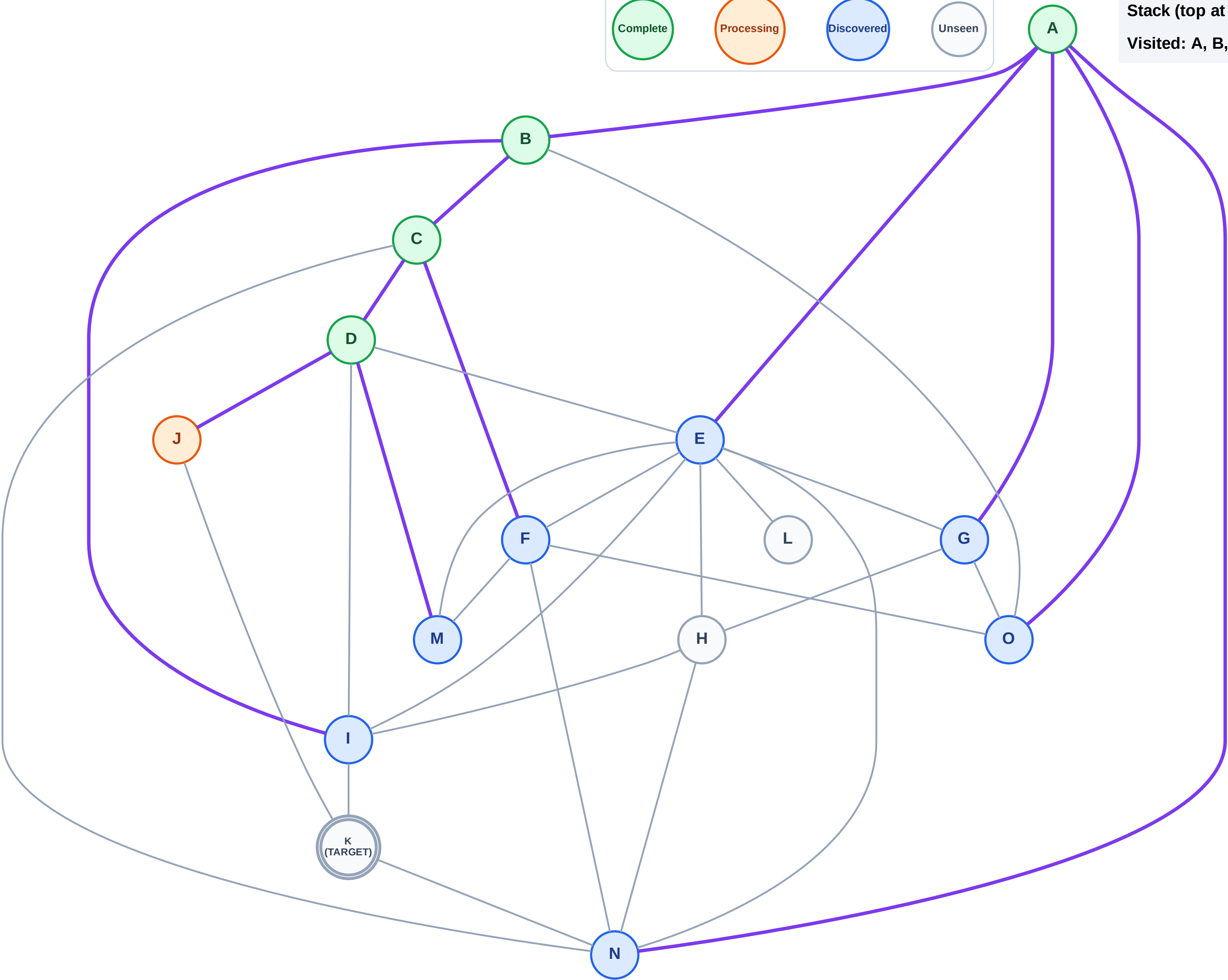
Processing

Discovered

Unseen

Stack (top at right): O | N | G | E | I | F | M

Visited: A, B, C, D



DFS Traversal

Step 11: Discover K from J

Legend

Complete

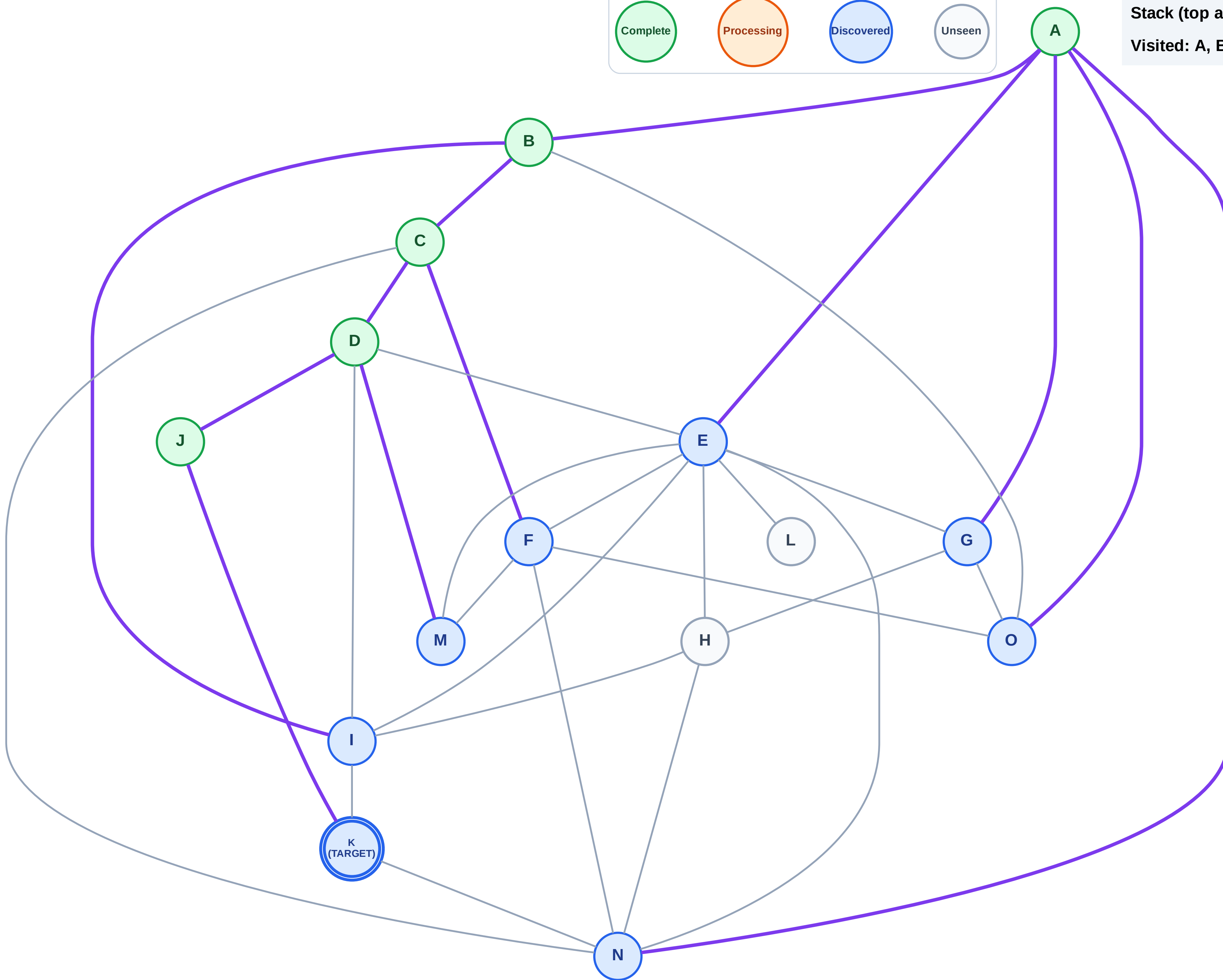
Processing

Discovered

Unseen

Stack (top at right): O | N | G | E | I | F | M | K

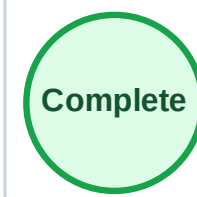
Visited: A, B, C, D, J



DFS Traversal

Step 12: Process node K

Legend



Complete



Processing



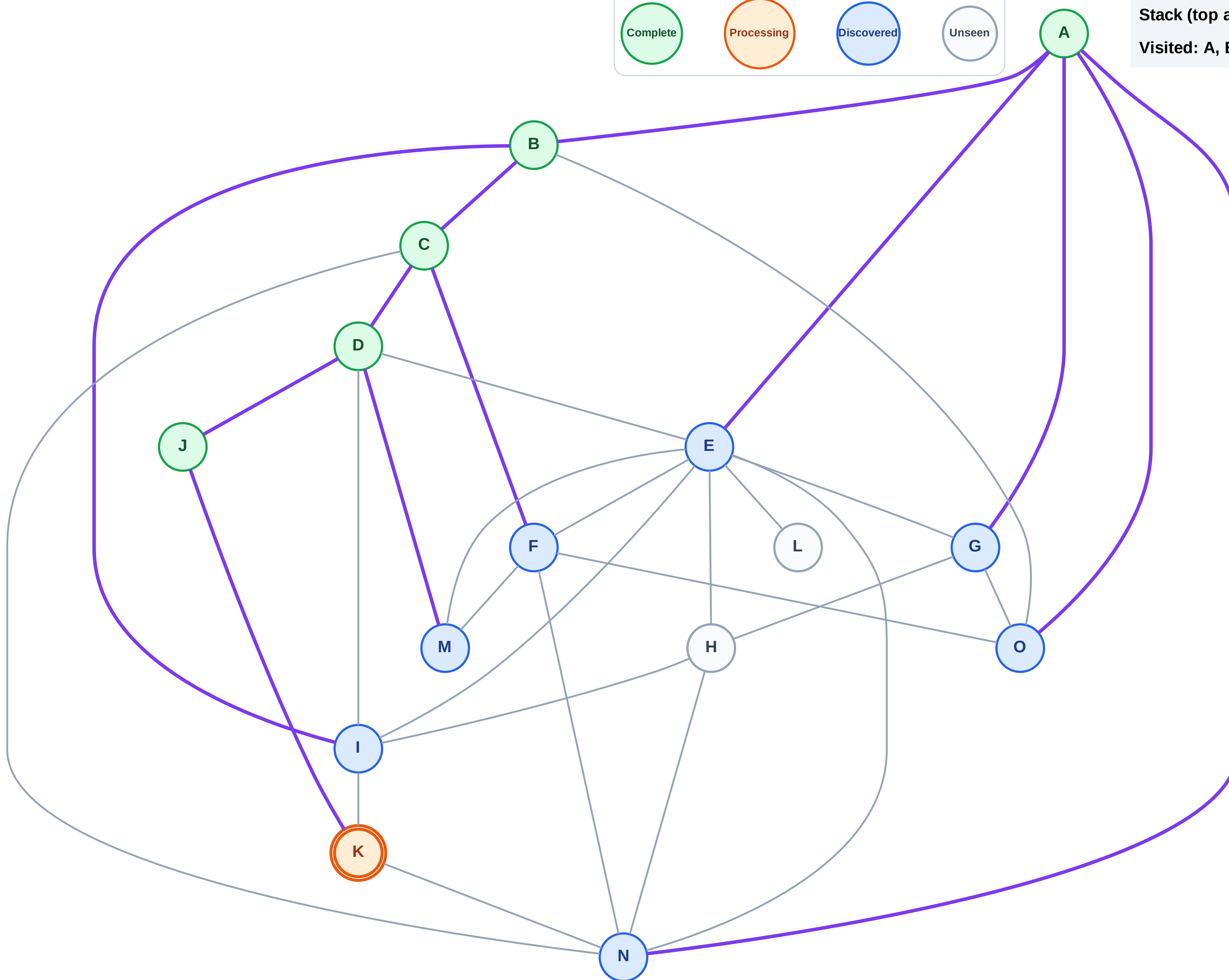
Discovered



Unseen

Stack (top at right): O | N | G | E | I | F | M

Visited: A, B, C, D, J



DFS Traversal

Step 13: TARGET NODE K REACHED! Path (5 hops): A → B → C → D → J → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | G | E | I | F | M

Visited: A, B, C, D, J, K

